Pre-Field Relativistic Geometric-Scaling Quantum Darkmatter

Topological Cosmic & Quantum Shortcut - Primary States of Field Generators

Bridging Classical & Modern Physics through single Ontology & Mechanism

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I) Abstract :

This Framework reinterpreting formation & survival of the matter in Big Bang without relying in problematic & paradoxical asymmetric CKM matrix model .

This Framework reinterprets Universe expansion as Global Contraction & Local expansion & The universe is tiny fraction on the surface of Incalculable Global Universe Sphere .

This Framework reinterprets Dark Energy , Darkmatter

This Framework modeling spacetime as a dynamic 4 dimensional geometrical description governed by Universe's components (Baryons , Darkmatter , Dark Energy , Radiation) .

This Framework reinterpreting Blackhole as a (Spatial Topological Shortcut) instead of Singularity , determining inlet & outlet area of Shortcuts by Surface tension .

This Framework modeling Blackhole in Quantum level , reinterprets multiple quantum behavior , reinterpreting quantum particles itself , including physical nature , reinterpreting their wave function origin , interaction with surrounding , mass , wave nature , charge , entanglement , superfluidity , superconductivity , magnetism , decay & redescriptions of particles .

The framework resolves cross-scale paradoxes and yields exact, non-adjustable quantitative predictions - without violating Michelson & Morley experimental findings .

Key verifiable derivations and forward predictions include:

Proton Charge Radius: A purely geometric derivation yielding 0.8412 fm, achieving 99.9% accuracy against the CODATA 2018 experimental value (0.8409 fm).

Cosmological Parameters: Mathematical deduction of the Universal Matter percentile at ~31.8% and Baryonic Matter at ~4.90%.
Black Hole Dynamics Defining black holes as topological void shortcuts (Confined Extra-Universal Void/CEUV/Void Bubble) rather than singularities, ..

Astrophysical Observations: Void Bubble radii predictions that precisely align with observed Blueshift spikes near M87 (predicting 5664 LY) and cooling filaments near Sagittarius A* (predicting 149.9 LY).

Mathematical Discovery of Hadron stars to substitute traditional phenomenon of Neutron stars (as mixture of neutrons & tetra-quark light mass sigma mesons content) Predicting a specific, measurable minimum & maximum mass limit of neutron/hadron stars yields at (1.22 & 2.37 solar mass) respectively , While White Dwarf Ultimate Mass limit at 1.52 Solar Mass

Quantum Entanglement Efficiency Equation in relation with temperature

This Paper also presents the following :

-Different Blackholes Annihilation Scenarios & The Actual Age formulae of Blackholes & its relation to Atomic Stability .

- Magnitude function of retro energy/massive matter creation .

- Speed of light constancy reason .

- Clarifying Fields Generators .

- Resolving Hubble Expansion rate Crisis preserving Quantum info conservation

- Clarifying Past Time Dragging (PTD) phenomenon & the extreme application of time manipulation & conserving quantum information .

The framework operates on strict mathematical parameters and offers direct observational points for falsification .

II) Keywords :

“Capillary” , “Capillary Wave” , “Absolute Capillary Wave Speed” , “Center of Percussion” , “Blackhole” , “EUV” , “CEUV” , “Void” , “Bubble” , “Void Bubble” , “VBH” , “Virtual Blackhole” , “Entanglement” , “Surface Tension” , “Corrected Planck Mass” , “Octants” , “Stucks” , “Chandrasekhar” , “Chandrasekhar Limit” , “Hadron” , “Hadron Star” , “Meson” , “Sigma Meson” , “Entropy” , “Local Entropy” , “Universal Absolute Capillary Wave” , “UACW” , ”Past Time Dragging” ,“PTD” , “Field” ,”Higgs Bosons” , “Bosons” , “W Bosons” , “Z Bosons” , “Neutrino” , “Casimir Effect” , “Annihilation” , “Proton Charge Radius” , “Electron” ,”Neutron” , ”Redshift” .

III) Principles :

The Proposed framework is based on the following principles :

1 – Capillary Wave Speed formula can be applied both in Newtonian & Relativistic formulation without relying on Relativistic mathematical correction

2 - Gravity is not an independent field , it is the interaction between Mass (which is physically equivalent to Energy) & Spacetime ... Actually it is the effect of Energy/Mass on a Spacetime … Energy/Mass is direct responsible for Gravity occurrence , As Gravity is the Curvature magnitude of Spacetime formed by Energy/Mass .

3 - Electrons size is not correlated directly to its wavelength or wave amplitude , it is the statistical probability of where that certain electron reside (Hence verified Coulombs Scattering study determined electrons as (point charges)) .

4 - Suggested Model does comply to all Quantum Mechanics & General Relativity verified observations & proven laws , as suggested model is an interpretative substrate model explains & resolve occurrences of point of clash , ambiguities in Copenhagen interpretation & General Relativity failing to reconcile at extreme states & Paradoxes , as it assume that QM & GR were accurate physical description of narrow domains .

5 - Hubble Expansion Rate is an effect of Dark Energy , Matter Density & Radiation which equivalent to ≈ 67.4 km/sec/Megaparsec = 2.184285 * (10^-18) m/sec/m as a (datum) according to CMB observation (The proposed model will clarify using this magnitude) .
----------(1)

6 - Elementary Particles , such as Electrons & Quarks , are considered (point charge) - dimensionless - respecting Coulombs Scattering verified results . (2)

7 - Proposed Model is Substrate framework of GR & QM both , & doesn't neglect them .

8 - Proposed Framework is dealing with darkmatter hypothetical properties based on potential energy & reinterpreting Darkmatter characteristics further in the framework .

9 -This framework posits that General Relativity & Quantum Mechanics are an emergent dynamic descriptions of a deeper existence governed by geometrical distribution of Spacetime & mechanical characteristics of Universe components . Therefore, while GR tensors Quantum field theory describe the macroscopic geometry, the fundamental physics is governed by the

constitutive equations of the proposed medium of Universal matter itself (Baryons & Darkmatter of Density and Surface Tension)

& this approach doesn't nullify Michelson & Morley experiment findings of Speed of light constancy relative to all frames of reference , nor reviving the idea of the old description of Ether as it was 300 years & lasted 2 centuries before Michelson & Morley experiment rule it out .

10 - The manuscript should be evaluated not on conformity with prevailing ontologies, but on internal consistency, parsimony, and testability.
By these criteria, it merits serious consideration and observational scrutiny rather than dismissal.

IV) Aims of The Framework :

1 - Solving The Loss of information inside a Blackhole .

2 - Model applied in both cosmic & Quantum world .

3 - Interpreting Probabilistic nature of Quantum Occurrences & reasoning constant parameters of cosmological Occurrences , as a scientific necessary tools used in serious attempt in a way to make Grand Unified Theory Possible , all the way down to TOE (Theory of Everything)

4 - Solving Hubble Expansion Rate magnitude crisis

5 - Reinterpreting Universal Creation without relying in Free Parameters , or Assumption of Unknown interactions

6 - Reinterpretation of Hawking Radiation & deduction of its effect on blackholes

7 - Reinterpretation of Neutron Star formation , source of rotation & prediction of mass limits

8 - Reinterpretation of Elementary fundamental particles

9 - Explanation of Mass/Energy reverse transformation

10-Fields Generation & Origins

11-Reinterpretation of Bosons nature

V) Postulates of Proposed Framework :

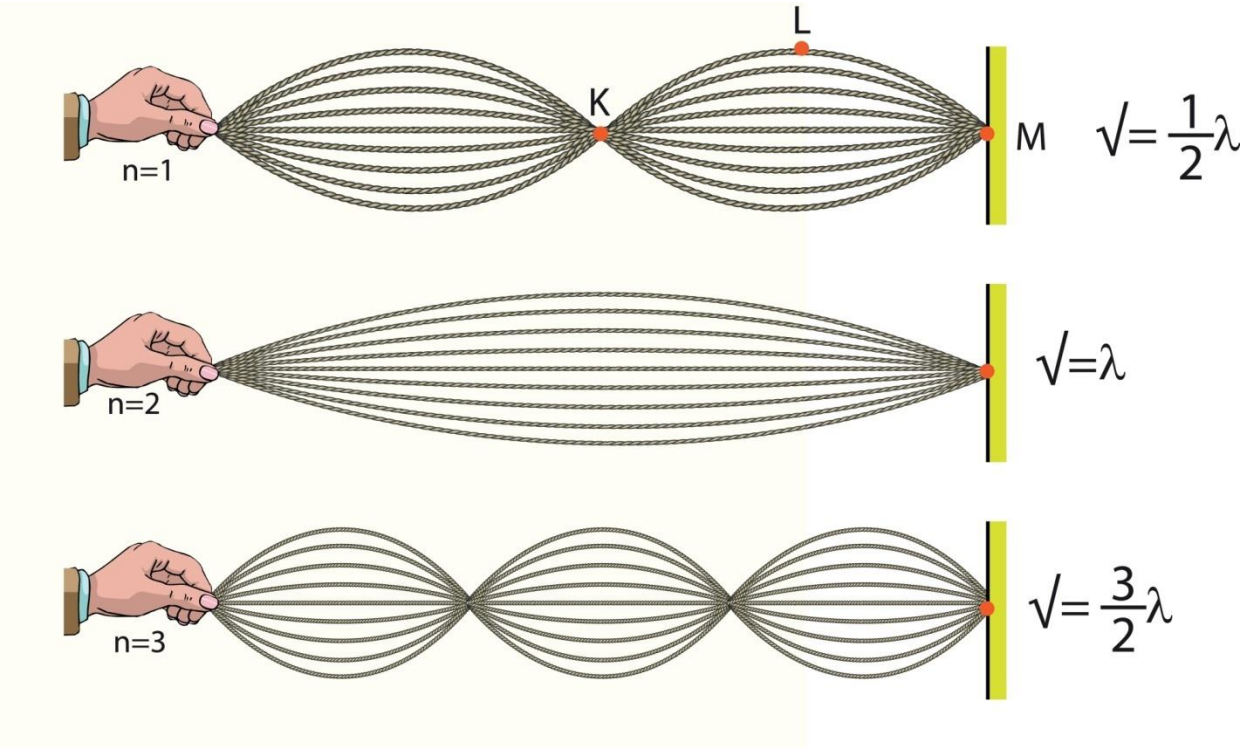
1 - Absolute Capillary Wave Speed is a Properties of The Object of Surface Tension σ & Density ρ when its Wavelength is twice third of its length for solid matters

As

$$V_O = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} \quad Where\ \lambda\ is\ wavelength\ at\ center\ of\ percussion\ (achieving\ resonance\ for\ solid\ states$$

$$V_O = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} \qquad Where\ (\lambda) = 2r/3 \quad (3)$$

Which 2r/3 is Center of Percussion



2 - Darkmatter doesn't interact with any baryonic matter physically or chemically except for gravitational effect that is unnoticeable locally , it is only noticeable in intergalactic level forming scaffolding of galactic clusters (Halos) . (4)(5)

3 - Dark Energy is the Net Universal Total Energy Released After Consumption of Formation of Dark Matter , Baryons & Radiation & Neutrinos , Until The framework describe the nature & cause of dark energy existence .

4 - Darkmatter (DM) +Baryonic Matter (BM) = Universal Matter (UM) .

5 - If the Framework propose that Speed of Light is The Absolute Capillary Wave Speed of The Universe (considering Universe has solid matter characteristics as it will be clarified in next chapters) , there's no correlation between light wavelength & Universal capillary wave Wavelength & that will be clarified in next chapters .

6 - Darkmatter physical properties are constant at all scales , hence its neutrality & extreme weak interactions with any substance , barely noticeable in Gravitational effect , in Halos .

So properties are constant , cosmic & microscopic , & this constancy is applied & conserved in the proposed model .

VI) Creation of Universe & Baryons (Corrected) - Qualitative depends on Model findings :

Standard Model Problems

1) According to Standard Model , at The first Second of The Universe , Quantum Fluctuations occur due to Complex Phase in Quarks Quantum State Transformations (as described in CKM Matrix) . (6)

While that considered a perfect resolution of Anti-symmetricity , but , it is conceptual not actual , hence the resulted percentage of Anti-Symmetric transformation won't match Baryons existed today , as for $(\eta_{CKM} \approx 10^{-19})$, $While\ Observed\ Baryon\ to\ Photon\ Ratio$ $(\eta_{obs} = 6.1 * 10^{-10})$.

Also , considering Dark Energy as vacuum energy is also challenging to be proven , hence the difference of energy magnitude differs from CMB observation by 100 power of magnitude

These extreme difference in magnitude represents paradoxes between what is been predicted theoretically , & what has been in observed experimentally & astronomically .

Assuming that Energy came out of Singularity at Planck Era of Planck Energy on a Planck Mass & Planck Length is also paradoxical .

Standard Model is ignores that Universe Total Energy is 2.64*10^71 joules , Dark Energy represents 68.16% & Radiation 0.009% , which (Zero Energy Universe Theorem Hypothesis) violates findings radically , violates these symmetricity entirely .

Zero Energy Universe suggesting neutralizing of Dark Energy by Energy Reduction of Redshift , while it is ignoring that red-shift affecting & manipulating the density of Energy , not the Energy existence from the start , it's not an energy created from nothing , energy transformations are magnitudes , not just resultant vectors .

So that refutes the claim of Big Bang creation from nothing automatically without energy needed , that violates energy conservation principle

Local Expansion /Global Contraction (LEGC) Mechanism

2) The Proposed Model interprets existence of Anti-Matter without total Annihilation , being generated & survived of the same magnitude as ordinary standard matter .

As Big Bang Blast Energy was stronger than total annihilation by interaction of Matter & Anti-Matter .

Inflation rate caused by big bang prevented Both particles from Total Annihilation .

Universal Inflation That Prevents Matter Annihilation resulted in symmetrical Distribution of Matter & Anti-Matter across Flat Universe , so Anti-Matter existed by the same magnitude as Matter , But further than to be Observed

Proposed model suggests That when inflation stopped , Universe is 4 dimensional sphere , Observed Universe is small patch on universe spherical surface area , Observed Universe regarded a (point) on spherical greater immeasurable Universal sphere , that explains flatness of observable universe , as Spherical global Universe contracts , curvature of the surface increase , makes local observed universe (which is considered a (point)) appear to be expands .

Dark Energy Expansion is actually The Attraction Between Matter & Anti-Matter

Dark Matter represents that Quantum Fluctuations (Interaction) between Both counterparts as Universe inflates , that generates Energy , so Dark Matter is just a pure energy resist Universe Local expansion & global contraction , same as potential energies of baryons
That Quantum Fluctuations energy distribution constitutes the scaffolding of later-born intergalactic clusters as the Universe rapidly inflates .

While Dark Energy is the display of Attraction of (Matter/Anti-Matter Counterparts) , Main Contributor to Universe Local Expansion & Global Contraction .

While Baryonic Acoustic Oscillations (BAO) dictates the formation of galaxies as temperature decreases due to Local Expansion/Global Contraction , meaning Our Observable Universe considered singularity at the surface of the whole Universe out of reach

So as The (Observable) Universe is (expanding) , The (Whole) Universe is Actually (contracting).

Perception of Decelerating Expansion Rate

Perception explains without free unknown parameters The Observable Universe Flatness which is proven from cold & hotspots size Observations of The CMB radiations , while the big bang resulted in a spherical inflation of Universe , explains Universe creation without relying on traditional standard model interpretation that suffers the difference in magnitude between theoretical calculated baryons to photons ratio due to CKM matrix to process survival of matter on the expense of antimatter by 9 power magnitude & dark energy & vacuum energy difference of near 100 power magnitude .

& While Flatness of Universe cannot be reconciled with Big Bang geometrically without any additional parameters unless one condition is fulfilled … That Observable Universe is just near infinite fraction of surface layer of a much more bigger Global Universe sphere (lead to a mathematical point on surface) , & That Universe reaches its maximum size through inflation due to Big Bang , & now Universe is shrinking back (dying) , & because Observable Universe is considered point on Universe surface layer , so as Universe contracts , curvature increases , angle of curvature increases , so Observable Universe (point Universe) expands & it is expanding at geometric rate as stellar objects are further from Observing point .

That mathematical perception also interprets decelerating Hubble Expansion rate

A Sphere that contracts at accelerating rate with mathematical point on the surface , as the surface curvature increases due to contraction , the mathematical point expands relative to the surface

But if the sphere contracts on accelerating rate , mathematical point expands in a decelerating rate due to following reasons :

1- Metric freeze

Surface containing the mathematical point contracts & runs out , & at singularity , dimensions = 0 , so there's no available dimensional capability for the point to expands & diminishes all local degree of freedom

2- Dimensional Collapse

As the mathematical point has no capacity to extend geometrically , local expansion is the differentiation the ratio between mathematical point volume & whole sphere volume , until

the sphere contracts in a singularity , meaning the ratio between mathematical point & singularity reaches maximum limit 100% (1/1) , so the differentiation will lead to zero

3- Time Dilation .. increased curvature will lead to time dilation as increasing curvature to singularity extends time infinitely .

VII) Framework Layout :

A) Universal

1 - Concluding Universal Matter Percentile Mathematically

Velocity Limit in any medium in space : V_0

$$V_O = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} \qquad Where\ V_0 = c$$

$$c = \sqrt{\frac{2\pi\sigma_{UM}}{\lambda\rho}} \qquad Where\ \sigma_{UM} = \frac{E_{UM}}{A}$$

Where $\lambda = \frac{2}{3}D_c$ (Assuming Solid State characteristics of The Universe)

Where $D_c\ is\ Universal\ Comoving\ Distance$

$$c = \sqrt{\frac{2\pi * (um) * \frac{4}{3}\pi * {D_c}^3\ \rho * c^2}{\frac{2}{3}D_c * \rho * 4\pi * {D_c}^2}}$$

After abbreviation

(um)= $1/\pi \approx 0.31830988618$

Where (m) is Universal Matter Ratio

These combination of solid & fluid characteristics of Universe & its geometrical shape will be studied , discussed & further clarified in next chapters & sections

(There will be clarification & distinction between universal absolute capillary wave & the light wavelength & frequency , that will be discussed in the following Conclusion section)

2 - Concluding Baryonic Matter Theoretically

First : Determining Ω_{BM}

From BBN Mathematical Proof (7)

1. The Physical Proof: The Deuterium Bottleneck

To understand why the amount of baryonic matter is fixed, we must look at the first few minutes of the universe (Big Bang Nucleosynthesis).

The Process:

1. Creation: Protons (p) and neutrons (n) collide to form Deuterium (D):

P + n = D + ɣ

2. **Destruction (The Bottleneck):** In the very early, hot universe, high-energy photons (γ) immediately blast Deuterium back apart. Deuterium cannot accumulate until the universe cools sufficiently.

3. **Fusion (The Drain):** Once Deuterium survives, it is immediately used as fuel to create Helium-4

D + D = He_4

The "Baryons" Argument:

- **If the Universe were dense (High Baryon %):** Collisions would be frequent. As soon as Deuterium formed, it would find a partner and fuse into Helium. The process would be extremely efficient, leaving **almost zero** leftover Deuterium.
- **If the Universe were empty (Low Baryon %):** Collisions would be rare. Nuclei would struggle to find partners. Much of the Deuterium would fail to fuse into Helium and would remain as "leftovers."

The Observation:

We observe a primordial Deuterium abundance of roughly

2.5 * 10^-5 (about 25 Deuterium atoms for every million Hydrogen atoms).

Physically, this is a "large" amount. It proves the early universe was not dense enough to burn all the Deuterium. This specific "inefficiency" locks the baryon density to a low number (~5%). If the density were even slightly higher (say, 10%), the Deuterium abundance would be virtually zero.

Mathematical Proof :

Using Precise Spectroscopic measurements from Quasar Absorption Spectra

D/H ≈ 2.53*10^-5

Plugging into BBN Standard Model yield to precise Power Law

$$\eta = 6.1 * 10^{-10}$$

Where

D/H $\alpha\ \eta^{-1.6}$

Converting Power Law (η) to Mass Density (Ω)

Calculating Photon density (np) using derived Power Law

$n_p = 2\frac{2\xi 3}{\Pi^2} * \left(\frac{K_B T}{h_{bar} c}\right)^3 = 0.411\ cm^3$

Barion Density (n_b)

$n_b = n_p * \eta = 2.5 * 10^{-7}\ bar$

Barion Mass Density (ρ_{BM})

$$\rho_{BM} \approx (2.5 * 10^{-7}) * (1.67492749804 * 10^{-24}) = \ 4.1873187451 * 10^{-31}\ gm/cm^3$$

Where $H_0\ is\ Hubble\ Expansion\ Rate = \frac{67.4\frac{km}{sec}}{megapasec}$

= 2.184285* 10^-18 Hz

h = 0.674

h^2 = 0.454276

From Friedmann Equation

$$\rho_{critical} = 3H_0^2/8\pi G$$

Where G is Gravitational Constant = 6.6743*10^-11 m^3 kg^{-1} sec^{-2}

$$\rho_{critical} = 8.5383542 * \frac{10^{-27} kg}{m^3}$$

Divided by h^2

$$\rho_{critical} = 1.8795521 \times 10^{-}29\, h^2 \frac{g}{cm^3}$$

$\Omega_{BM} = \frac{\rho_{BM}}{\rho_{critical}}$ $=(4.1873187451 * 10^{-31})/(1.8795521 \times 10^{-29})$ =0.02227827973

(bm) = $\frac{\Omega_{BM}}{h^2}$ = 0.02227827973/0.454276 = 0.04904128708= 4.904128708 % of Total Universal Energy (UE) (8)

This is determination of BM ratio out of BBN model Mathematical methodology & proof , it is not the framework invention

3 - Concluding DarkMatter ratio (dm) :

(dm) = $0.31830988618 - 0.04904128708 = 0.2692685991 = 26.92685991\ \%$

4 - Concluding Dark Energy Ratio (de)

Where Radiation Percentile = 0.00009 (9)

(de) = 1 - 0.31830988618 - 0.00009 = 0.68160011382 = 68.160011382 %

5 - Concluding Comoving Distance - Volume - Surface Area - Mass - Energy & Energy Equivalent Distribution :

From Friedmann Formula (8)

$$D_c = \frac{299792458}{2.184285 * 10\text{^} - 18} * \int_0^{\infty} \frac{dz}{\sqrt{(r)(1+z)^4 + (um)(1+z)^3 + (cr)(1+z)^2 + (de)}}$$

Where (r) is radiation ratio

Where Curvature Parameter (Cr) = 0

Hence Universe is flat

$$D_c = \frac{299792458}{2.184285 * 10^{-18}} * \int_0^{\infty} \frac{dz}{\sqrt{(0.00009)(1+z)^4 + \left(\frac{1}{\pi}\right)(1+z)^3 + (0)(1+z)^2 + (0.68160011382)}}$$

= 4.34740212246533915608085025535* 10^26 m = 45.98355071799456701427 billion light years = 14088.97076137215 megaparsecs

V = 4/3*π*(4.34740212246533915608085025535*10^26)^3 = 3.44173991488*10^80m3

Surface Area = 4*π*(4.34740212246533915608085025535*10^26)^2 = 2.37503213501*10^54 m2

M = $\rho_{critical} * V = 8.5383542 * 10^{-27} * 3.44173991488 * 10^{80}$

M = 2.9386794458*10^54 kg

Total Energy (TE) = 2.9386794458*10^54 * 299792458^2 = 2.6411533706*10^71 joules

Universal Matter Energy Equivalent = $\frac{1}{\pi}$ * (TE) = 8.4070522879*10^70 joules

Darkmatter Energy Equivalent = 0.2692685991 * (TE) = 7.1117966811*10^70 joules

UM Surface Tension = (8.4070522879*10^70)/(2.37503213501*10^54) = 3.539763594762730384720614 61*10^16 Kg/sec2 (j/m2)

DM Surface Tension = (7.1117966811*10^70) / (2.37503213501*10^54) = 2.994400192008372972216612248*10^16 Kg/sec2 (j/m2)

Universal Age = t© =

$$\frac{1}{2.184285 * 10^{-18}} * \int_0^{1} \frac{dz}{z\sqrt{(0.00009)(z)^{-4} + \left(\frac{1}{\pi}\right)(z)^{-3} + (0)(z)^{-2} + (0.68160011382)}}$$

= 4.339258564603238588371023012 1 * 10^17 sec

= 13.75969864473*10^9 years (13.7596986 billion years , just slightly younger than latest predictions of 13.8 bil)

Hubble Expansion Rate @ Redshift Magnitude of (z=1100)

$$H_z = H_0 \sqrt{(0.00009)(1+z)^4 + \left(\frac{1}{\pi}\right)(1+z)^3 + (0)(1+z)^2 + (0.68160011382)}$$

$$H_{1100} = 2.184285 * 10^{-18} * \sqrt{(0.00009)(1+1100)^4 + \left(\frac{1}{\pi}\right)(1+1100)^3 + (0)(1+z)^2 + (0.68160011382)} =$$

$$5.15544840973 * 10^{-14}\, Hz = 1590805333.62622988872 \text{ m/sec/mpc}$$

= 1,590,805.33362622989 km/sec/megaparsec

Universal Age at Redshift magnitude of z=1100

$$\frac{1}{2.184285 * 10^{-18}} * \int_{1100}^{\infty} \frac{dz}{(1+z)\sqrt{(0.00009)(1+z)^4 + \left(\frac{1}{\pi}\right)(1+z)^3 + (0)(1+z)^2 + (0.68160011382)}}$$

= 11,543,396,048,729.9047514404 sec ≈ 366,039 years

According to current Observation , there were contradictions between measuring Hubble Expansion rate parameter using observed redshift of nearby & far old galaxies (Standard Candle methodology) as it observed expansion rate magnitude locked at 73 km/sec/mpc , & Calculating expansion rate using Einstein metric tensors & Friedmann formulae according to CMB radiation observed to lock expansion rate mathematically magnitude of 67.4 km/sec/mpc.

According to the proposed model explanation , CMB expansion rate deduced is the (Current) - Local - expansion rate of the Observed Universe

& the expansion rate of magnitude of 73 km/sec/mpc observed using Standard Candle method is The Arithmetic mean of expansion rate through time , & direct observation of galaxies redshifts (even near galaxies & stars) cannot be got rid of time dimension & time dilation effect , & it is not about observing galaxies itself only , it is about observation of the expansion rate evolution itself .

& According to Proposed Model perception of Universe creation , expansion rate magnitude of 73 km/sec/mpc is not only arithmetic mean of expansion since Light liberation at 366,039 years , it is the arithmetic mean of the sum of magnitudes of (local contraction rate) accompanied by global inflation & (local expansion rate) accompanied by global contraction of interaction of attraction between matter & antimatter across universe .

6 - Expansion Rate Lower limit & Proposed Model definition

Lower Limit of Expansion rate is the limit where expansion decelerating rate will equals = 0

It is the same limit where redshift will equals = -1

Where

$H_z = H_0\sqrt{0.68160011382}$ = $2.184285 * 10^{-18} * \sqrt{0.68160011382}$

$= 1.80332552752 * 10^{-18} Hz = 55.64481766567 \frac{\frac{km}{sec}}{mpc}$

Universe Age at Lower Expansion rate limit (t)

=

$$\frac{1}{1.80332552752 * 10^{-18}} * \int_0^{\infty} \frac{dz}{(1+z)\sqrt{(0.00009)(1+z)^4 + \left(\frac{1}{\pi}\right)(1+z)^3 + (0)(1+z)^2 + (0.68160011382)}}$$

= 5.25594367136759889657395648 56*10^17 sec = 16,666,488,049.74504977351 years = 16.67 billion years

According to proposed model

the expansion rate lower limit doesn't mean that the universe will be at steady state expansion (as some models predict within Standard model domain) , but according to proposed model this is the lower limit of (local) expansion rate while the Observed universe acting as (flat) .

After that curvature ratio (Cr) will be positive (Due to global contraction) on the expense of Dark energy parameter , & local expansion will be accelerated (since Curvature has (1+z)^2 redshift parameter)

Also , reaching Lower limit expansion rate in 2.91 billion years from now , is the maximum duration it will take to reach that magnitude

According to the proposed model
The Universe contracts due to attraction between matter & antimatter , in case of first matter & antimatter interaction , it will lead to increase of radiation (r) parameter in expense of Universal matter ratio (um)

This will lead to local expansion accelerated rate (Since radiation parameter (1+z)^4 , while Universal matter has only redshift parameter of (1+z)^3)

& then The observable universe will reach flat limit sooner .

B) Beyond Universal

(Observable) Universe Expansion Speed surpassing of Spacetime Speedlimit :

Assuming Availability of Observing furthest Object at 4.347402122* 10^26 m = 45.98355 billion light years = 14088.97076137215 megaparsec

Hubble Expansion Rate is 67.4 km/sec/megaparsec = 67400m/sec/megaparsec

Object Speed hypothetically appeared = 67400*14088.97076137215 = 949,596,629 m/sec (more than 3 times speed of light)

But This speed is achieved in (Observable Universe) …. Rest of the universe expansion speed at the limit is unknown to us

But Outside of The Universe is nothing , void (emptiness) , no mass , no density , no dimension , no spacetime - not physically determined in Standard Model terms

That Extra-Universal-Void is regarded by Standard Model term as Non-Physical Void , Which is different from Traditional-Physical-Vacuum (which Standard Model is dealing with) which filled by virtual particles which causes Quantum Fluctuations as it had been noted in Casimir effect , This Physical Vacuum will be described later further in the framework .

According to The Proposed Model , The Extra-Universal-Void (EUV) is what Standard model would describe as Non Physical - Proposed Model would Describe as EUV (Extra-Universal-Void) , which is (the emptiness) outside the universe as immeasurable undetermined void .

Applying absolute capillary wave Velocity limit of that Void Outside The Universe (EUV)

$$V_O = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} V_O = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} = \frac{0}{0} = indeterminate\ value$$

Meaning inability to determine velocity (speed) limit there … Actually (Velocity) is (indeterminate) Value , this result would benefit in studying microscopic world & quantum level analysis .

C) Cosmic :

1) Studying Blackholes

Proposed Model suggested that Blackhole formation limit is evenly distributed across all baryons of stellar object (Neutron star) , but that interpretation is not inclined with decreasing density as blackholes get bigger , that paradox exist if we consider singularity formation as Standard model assume (the alternative assumption will be clarified further in the framework)

1 - Proposed Model Interprets blackhole limit as the amount of mass accumulated enough to transform every neutron of the spherical object to a micro blackhole , giving the blackhole formula :

$$R = \frac{2Gm}{c^2} \qquad (10)$$

So $m = R(\lambda) * \frac{c^2}{2G}$

As

$\lambda = Planck\ Length = 1.616255 * 10\text{^} - 35\ m$ (11)

$c = 299792458\ m/sec$

$G = 6.6743 * 10^{-11}\ m\text{^}3/(kg.sec\text{^}2)$ (12)

$$m = (1.616255 * 10^{-35}) * \frac{(299792458)^2}{2 * (6.6743 * 10^{-11})} = 1.088217 * 10^{-8} kg$$

(that's precise exact half of what Planck deduce mathematically)

this is the wave length of maximum kinetic energy neutron could offer to degenerate gravitational pressure before collapsing by its mass

so Planck Mass formula should be = $0.5\sqrt{\frac{hc}{2\pi G}}$ instead of $\sqrt{\frac{hc}{2\pi G}}$ (13)

& that means correction of Kinetic Energy formula for nucleon

Where at maximum stability for Neutron before collapse

P.E = K.E

From De Broglie Wave Formula $\mathrm{mc}^2 = \frac{(\mathrm{K})\mathrm{h}^2}{\mathrm{m.l}^2}$ (14)

where (l) is Planck length , equivalent to (Rs)

$\mathrm{L_{Planck}} = 2\mathrm{G.m/c}^2$

$m = \mathrm{L_{Planck}} * \frac{c^2}{2G}$

apply in equation

$\mathrm{L_{Planck}} * \frac{c^4}{2G} = (K) * \frac{2G*h^2}{c^2*l^3}$

$\mathrm{L_{Planck}}^4 = (K) * 4 * G^2 * \frac{h^2}{c^6}$

since (l) is Planck length = 1.616255*10^-35 m

G = 6.6743*10^-11 m3/(kg.sec^2) (11)

C=299792458 m/sec

(h is Planck constant = 6.62607015*10^-34 j.sec (13)

(K) is the corrected constant of De Broglie Wave Formula

after calculation

K= $1/16\pi^2$

This is the value of kinetic energy constant that align with the fact that maximum kinetic energy of a particle reflect into minimum wavelength (Planck Length) resulted into maximum potential energy for a neutron that is exact half of hypothetical Planck Mass to be equals 1.09*10^-8 kg (as Schwarzschild radius is 1.616255*10^-35 m) , hence no need to assume that Schwarzschild radius will equal 3.23251*10^-35 m (as twice of Planck length)

2 - The minimum mass limit that is enough to form neutrons of spherical stellar object in space before turning into a blackhole

From Schrodinger Formula: (15)

$$\Delta x * \Delta p = \frac{h}{2\pi}$$

Where

Δx = (R^3/n)^1/3 = R/(n^(1/3))

$$\Delta p = \frac{h_{bar}}{\Delta x} = h * \frac{n^{\frac{1}{3}}}{(2\pi) * R}$$

K.E = Δp*c = $h * \frac{n^{\frac{1}{3}}}{(2\pi)*R} * c$

P.E = Gravitational Energy = $\frac{GMm}{R} = G * (nm) * \frac{m}{R}$

P.E = K.E

$G * (nm) * \frac{m}{R} = h * \frac{n^{\frac{1}{3}}}{2*\pi*R}$ * c

$G * (nm) * m = h * \frac{n^{\frac{1}{3}}}{(2\pi\ )}$ * c

$G * \left(n\text{^}(\frac{2}{3})\right) * m\text{^}2 = h * \frac{c}{(\pi\ )}$

$\left(n\text{^}(\frac{2}{3})\right) = h * \frac{c}{(2\pi\ )*G*m\text{^}2}$

$$n = \left(\frac{\left((6.62607015*10^{-34})*(299792458)\right)}{(2*(\pi)*(6.6743*10^{-11})*((1.67492749804*10^{-27})^2))}\right)^{\frac{3}{2}} = 2.1940608727 * 10^{57} neutrons$$

Where for the Sun …. No of Solar Neutrons = Solar Mass/ Neutron Mass

= $1.98847 * 10^{30}$ / $1.67492749804 * 10^{-27}$= $1.18719765621 * 10^{57}$ neutrons

Theoretical Blackhole mass limit = 2.1940608727/1.18719765621 = 1.848100744828204785123057044785 9 Solar Mass

No need to consider Strong interaction in the equation to increase limit to 2.3 solar mass … hence the interaction occurred in neutron particle mass itself which includes strong interaction between its quarks (& strong interaction nature will be discussed further in the framework in microscopic model subsection)

But The **Actual** Blackhole formation mass limit differs in magnitude than 1.85 Solar mass hence there is another parameter come into consideration other than strong interaction & that is discussed in Neutron Star formation section according to the proposed model .

2) Modern theories struggled to explain occurrence of entanglement

Entangled particles transfer information to one another in no time (teleporting information) even if entangled particles were in opposite sides of the universe , & that contradicts with General Relativity .

Also Modern theories struggled to explain (mechanically) , how electrons elevated from Quantum state to another without passing through in between (The reason of Quantization).

Quantum Mechanics cannot explain (mechanically) the electron cloud occurrence

This framework started searching for answers cosmically , looking through a blackhole .

Blackholes generated through massive Star ... the gravity of that massive star are stabilized by the energy generated out of nuclear fusion occur in the star core due to gravity pressure itself ... until all nuclear fuel are wasted ... all of hydrogen are fused into helium , helium fused into carbon & oxygen ... etc. ... until it fused into Iron ... Iron atom is stable enough to be very resistant to Fusion to form any heavier element , the stellar object explode in a supernova .

According to Tollman-Oppenheimer-Volkoff (TOV) equation , if remnant of stellar object after the explosion of Supernova exceeds ≈ 2.3 Solar Masses (According to TOV model of course) , it contracts into Blackhole , Hence Pauli Exclusion Principle (PEP) for neutrons , cannot counteract excessive gravity pressure generated . (16)

The recent known phenomenon Lifetime of stars , that are formed as clouds of molecular hydrogen resulted out of inflation after the Big Bang , these clouds has potential energy curved spacetime (generating gravity) combining molecular Hydrogen through time , the accumulative energy through time affected hydrogen & forces it to fuse , generating stars as we know it , the stars fuses hydrogen as a result of their gravity creating nuclear fusion energy to counteract potential energy/mass/gravity … until star consumed its hydrogen fuel , &

resulted matters used as a fuel (Helium , Carbon , Oxygen & Silicon) for nuclear fusion , gravity pressing star matter , making extra nuclear fusion .

Until Gravity of Stellar Object mass cannot force resulted core matter to fuse , whether its 4 elements mentioned above , or final phase of nuclear fusion results …. Iron

The rebound wave will destroys all upper layers of The stellar object in a supernovae for massive star , or cosmic winds due to weak gravity for less massive star , leaving remnant of non fusable stellar object behind .

The remnant of the stellar object faces 3 scenarios

1-)Forming white dwarfs ... Electrons degeneracy pressure are stabilizing force of gravity due their PEP … that occur for less massive stars

2-)Forming neutron stars , PEP for electrons unable to counteract excessive gravity by stellar object remnant mass , electrons fall to interact with protons forming neutrons , & Pauli's Exclusion principle that prevents Quantum particles to share same quantum states , stabilizing force of gravity … that occur for intermediate massive star

3-)Forming blackhole , gravity field overcome Pauli's Exclusion principle for nucleus , General Relativity suggests that the matter will contract into singularity. Which Quantum Mechanics refuses that outcome that violated Quantum Information Conservation & also violate conservation of energy (as it will be explained later) .

Blackhole principle is that mass(energy) of the cosmic object bends Spacetime infinitely so nothing can escape from gravity , even Electromagnetic waves (such as light) cannot escape from .

According to General Relativity (GR) itself , that stellar object material when got turned into a blackhole , it will reach singularity , that singularity is dimensionless particle , which means it has an infinite density , while Quantum Mechanics (QM) refuses the concept of singularity , GR also ignores that infinite density means infinite mass , infinite mass meaning infinite energy , infinite mass happen also at speed of light , infinite energy violates conservation of energy .

3) Proposed theory suggested Blackhole formation mechanism

Potential Energy generated from The star , that generated energy explained by Einstein formula PE=M.C^2

that energy is being neutralized & manifested by Star core nuclear fusion of its hydrogen fuel , hydrogen fuel is being consumed throughout billions of years.

after hydrogen fuel being completely depleted out of the star , the star begin to manifest that energy by Helium nuclear fusion , helium fuel is being consumed & depleted , star grows larger as it fuses larger & heavier elements until it formed Iron element in its core , the star then will transform into Red Giant , Iron is very stable & harder element to be fused to heavier atoms , then suddenly the star explodes forming Supernova & remnant material of mass (m) of heavy elements of the star suffers inability to generate nuclear fusion , nothing could neutralize Potential Energy that forming that stress

Remnant of the star crush to its core with by its own weight (as TOV predicted mathematically) with the existence of resistance of the atoms to be crushed on itself .

Gravity overcoming PEP degeneracy pressure of neutrons if the mass of remnants exceeds 1.85 Solar mass - as we prove earlier (theoretically) - & Crushed the matter , Gravity Exceeds until light can’t reach its escape velocity , it can’t escape gravity , forming a blackhole .

4) Proposed Framework Model of Blackholes

Proposed model suggest that Gravity will push stellar matter to contract (in case of blackhole) that contraction & pressure increase neutrons kinetic energy as described earlier , where the neutrons to form a blackhole requires that its kinetic energy will reach magnitude as its neutron velocity will reach speed of light (so that explaining that electromagnetic wave cannot escape Blackhole Gravity) , that will lead to infinite mass according to General Relativity , so infinite Kinetic Energy , bend spacetime infinitely as it explodes Universal Matter

(This description is not totally correct , hence infinite Kinetic Energy will also violates conservation of energy , But The introduction of the precise description needed studying proposed model at quantum scale , The precise perception will be discussed further in conclusion section)

But , The Baryonic Matter is already collapsed due to its own mass (Potential Energy) , The Energy affect only the Dark matter (rest of universal matter) which lead to different thesis

that kinetic energy of collapsed pressurized neutrons exploding the Dark matter creating a (bubble of EUV) , that confined EUV is what the proposed framework call (Void bubble) or Confined/Contained EUV or CEUV ... a void in Dark matter , Void Bubble (or Sphere) in Universe itself .

, The dimension of Void Bubble determined mathematically from the former formula

$$\sigma_{DM} = \ 2.9944002 * 10^{16} \ kg/sec^2 \ or \ \ joules/m^2 = E/A$$

Where E is Potential Energy of The Blackhole

Where A is the surface area of spherical Confined EUV that has surface tension of $2.9944002 * 10^{16} \frac{kg}{sec^2}$

= Mc^2 / 4πRb^2

Where R_b(R Bubble = Radius of Void Bubble = Radius of CEUV) is The radius of Void Bubble (CEUV) created out of Black hole

This Case is different from EUV Containing the Universe , in this Case , The Universe containing that Void

& in this (contained) immeasurable void , matter accreted teleported in no time , as velocity in this vicinity (Void Bubble) is infinite according to the formula

$$V_O = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} \qquad Where\ (\sigma) = 2.9944002 * 10^{16}\ kg/sec^2 \qquad While\ \rho = zero$$

$$V_O = \infty$$

(That Spooky Action in no time that Einstein refused it)

& That doesn't mean that Object moves in CEUV faster than (c) (actually in infinite speed , which violates causality)

That CEUV is just confined EUV , but has same characteristics , it has no energy , no surface tension , no dimensions , but as being confined , it works as Topological shortcut , that bend spacetime to make a shortcut in 4 dimension spacetime

This result solving (Loss of Information) , as Quantum Mechanics (Information conservation) is Preserved .

& interpretation of infinite speed is crystal clear , that dimensionless void bubble perform as a Topological Shortcut , so infinite speed is justifiable hence un-locality .

5) Correlation Between R(bubble) & R'(Schwarzschield)

$$R'_{Schwarzschild} = \frac{2GM}{c^2}$$

$$R_{Bubble} = \sqrt{\frac{Mc^2}{4\pi\sigma}}$$

$$\frac{R_{Bubble}}{R'_{Schwarzschileds}} = \frac{c\text{^}3}{4G\sqrt{\pi M\sigma}}$$

According to Verified Experimental Observations

Blackholes gaining mass of matters being accreted or sucked into it

According to the proposed model . Inlet of the proposed Void bubble is correlated to the outlet (Border with Universe , earning additive Potential Energy from matter teleported through Void Bubble (CEUV) to it , that teleported matter will generate gravitational energy that will curve Universe in the outlet , that Curvature is actually a (Tensile stress/Energy) forcing the Outer Spacetime/Universe to shrink as forcing Void Bubble Sphere (CEUV) to expand

While the inner Universe containing the Blackhole itself expands by the same rate .

That expansion/contraction magnitude excess is equivalent to potential energy/mass of the object teleported (passing through) .

Which being translated as that (Blackholes earning mass of objects got accreted) , while the earning mass/energy is actually equivalent to the expansion of dimensions of Void Bubble , hence the constancy of Surface Tension of the surrounding Dark matter , which is translated in that formula :

$\Delta R = \sqrt{\Delta E/4\pi\sigma}$, $\Delta R^2 = \Delta E/4\pi\sigma$, $2RdR = dE/4\pi\sigma$, $RdR = dE/8\pi\sigma$

Where ΔR is change in radius of the Bubble , ΔE is potential energy increase , σ is the surface tension of Darkmatter (Constant) .

6) Active Blackhole Ultimate Growth & Annihilation - Collapse By Spatial Equilibrium

Case (A) According to The Proposed Model , Relation Between R_{Bubble} & $R_{Schwarzschild}$ is formulated in this derivation

$$\frac{R_{Bubble}}{R_{Schwarzschild}} = \frac{c^3}{4G\sqrt{\pi M\sigma}} = \frac{299792458^3}{4 * 6.6743 * 10^{-11}\sqrt{\pi * 2.9944002 * 10^{16} * M}}$$

That means as Blackhole accreting , Event Horizon expand faster than CEUV as

$R_{Schwarzschild}\ \alpha\ M$ while $R_{Bubble}\ \alpha\ \sqrt{M}$

Until Both Event Horizon & CEUV matches each other

Where $M = \frac{c^6}{16\pi\sigma * G^2} = \frac{299792458^6}{16\pi * (6.6743 * 10^{-11})^2 * 2.9944002 * 10^{16}}$

Case (A) Blackhole Mass = 1.08276216787*10^53 kg (Where Bubble/CEUV Radius & Schwarzschild Radius are matched)

$$R'_{Schwarzschileds} = \frac{2GM}{c^2} = \frac{2 * 6.6743 * 10^{-11} * 1.08276216787 * 10^{53}}{299792458^2}$$
$$= 1.60815307838819777461496095197 * 10^{26}\ m$$

As Light Year = 9.454254955488 * 10^15 m

So Schwarzschild Radius = 16.998191450955778572 65 Billion LYs

Which would exceed Observable Universe Age .

But this is completely wrong phenomenon , because inside Event Horizon according to Standard Model , Our Laws of Physics collapse .

& According to The Proposed Model , inside Event Horizon (Schwarzschild Sphere) is Contained (or Confined) Extra-Universal Void , which is Nothingness outside the universe , which is not bound by Universal Spacetime constraints , & that means that Schwarzschild radius here is magnitude of distance (from our universal perspective) , Otherwise The Event Horizon itself is just a 3D gate of a Contained Extra-Universal Void (CEUV) which extends beyond Event Horizon boundaries , especially in the birth & growth phase of blackhole .

But According to Proposed Model , Under accretion , The growth rate of Schwarzschild radius surpassing its counterpart of CEUV/Bubble Radius , as displayed from the derived equation above

Until both radii are matched

According to Proposed Model

Any accretion occurs beyond that limit , the accreted object teleported inside Event Horizon sphere & won't surpass it , it means that the Universal section outside the CEUV would reside inside Event Horizon sphere which will lead to violation of conservation of energy , which is not allowed according to proposed model

So The Proposed Model solved this problem by assumption of Blackhole collapse occurs at this equilibrium state

& Universe stretched to occupy space of The blackhole

& Due to Our very limited detection capability , that doesn't exceed 1% of The Observable Universe

So , The Correct statement is not that Blackhole limit would exceed Observable Universe Age

It is that blackhole reaching this limit is beyond our detection capability

Schwarzschild Radius limit is spatial magnitude other than chronic magnitude , hence inside blackhole is not bound spacetime geometry constraints according to both Standard model & this proposed model .

7) Annihilation by Dark Energy

According to Proposed Model , As assuming Blackhole as Geometric Topological Void at Universe Spacetime (Contained Extra Universal Void – CEUV) , As a result , CEUV should be shrinking over time as Universe locally expands (Globally contracts) by the local rate of 67.4 km/sec/megaparsec (= 2.184285 * 10^-18 m/sec/m) & accelerating over time according to the proposed model , Until it reaches Planck Length = 1.616255*10^-35 m

The Blackhole expands as it accretes & contracts as The Universe locally expands (globally contracts)

The Scenarios applied in this model , assuming The Blackhole is dormant (Such as System GAIA BH01)

There are two perception

A) Assuming Contraction rate of CEUV radius decreases locally by decreasing contraction rate Until it reaches Planck Length

Assuming Hubble Expansion Rate which equals = 2.184285 * 10^-18 m/sec/m as (K)

$$\frac{dR}{dt} = -K.R$$

$$\frac{dR}{R} = -K * dt$$

$$\int_{R_0}^{R_{Planck}} dR/R = -K \int_0^t dt$$

$$\ln R = -Kt + C$$

$$@\ t = 0$$
$$\ln(R_0) = C$$

$$\ln(R_{Planck}) - \ln(R_0) = \ln\left(\frac{R_{Planck}}{R_0}\right) = Kt$$

$$R_t = R_0 e^{-Kt}$$

$$@\ R = l_{Planck}$$

$$e^{-Kt} = \frac{l_{Planck}}{R_0}$$
$$e^{Kt} = \frac{R_0}{l_{Planck}}$$
$$Kt = \ln\left(\frac{R_0}{l_{Planck}}\right)$$
$$t = \frac{1}{K} * \ln\left(\frac{R_0}{l_{Planck}}\right)$$

B) Assuming Contraction rate of CEUV radius decreases locally by local constant contraction rate Until it reaches Planck Length

$$\frac{dR}{dt} = -K.R$$
$$\int_{R_0}^{R_{Planck}} dR = -\int_0^t (KR)dt$$
$$R(t) = -(K * R_0)t + C$$
$$@t = 0$$
$$R = R_0$$
$$C = 0 + R_0 = R_0$$
$$R(t) = R_0 - (K * R_0)t$$
$$R(t) = R_0(1 - Kt)$$
$$@R = l_{Planck}$$
$$l_{Planck} = R_0(1 - Kt)$$
$$t = \frac{1}{K}\left(1 - \frac{l_{Planck}}{R_0}\right)$$

While in Cosmic Range $\lim_{R\to\infty} \frac{l_{Planck}}{R_0} = 0$

$$t = \frac{1}{K} = \frac{1}{2.184285 * 10\text{^} - 18} = 4.5781572 * 10^{17} \text{ sec} = 14.517241103\ billion\ years$$

That is All Cosmic Blackholes estimated age , whatever their size & mass

8) Notes

A) The Framework did not discuss the effect of star-blackhole-Void bubble spinning , as thought to be has no significant value on determining Void Bubble radius , or Coordination of inlet/outlet of teleported object … since the kinetic energy of the star spinning is conserved

B) Another Aspect should be noted :

That Energy preservation occur as soon as Stellar Object collapse into a blackhole , where its matter (Energy) forming the CEUV .

C) CEUV Relativistic Compliance

CEUV Doesn't have any gravitational lensing or interacting with Observable world , since its nothing , un-local ... it doesn't affect any matter outside the proximity of its gate (Blackhole)

D) A potential critique of the proposed model is the assumption of dark matter homogeneity across the observable universe. Since primordial inhomogeneity is the established mechanism for the formation of intergalactic clusters & Galactic Halos , there would be an argument that this contradicts the assignment of a singular, universal magnitude to dark matter surface tension. Consequently, this perceived inhomogeneity could be viewed as incompatible with the model's mathematical mechanism for black hole formation, which fundamentally relies on universal dark matter homogeneity.

To address this, the proposed model postulates that dark matter inhomogeneity was strictly a transient phase during the early universe, immediately following the Big Bang (a concept elaborated in the subsequent sections Universe Creation and the subsequent section of Conclusions). This primordial inhomogeneity phase catalyzed the formation of both dense and light stellar structures.

Furthermore, this localized structural evolution is perfectly consistent with the Cosmic Microwave Background (CMB) and the Lambda-CDM model, which confirms that the universe—and by extension, dark matter—remains fundamentally homogeneous and isotropic at macroscopic scales exceeding hundreds of millions of light-years.

To Solve that Paradox , Proposed Model suggested that Galaxies (which its core formation conserved due to existence of central Supermassive Blackhole) deform spacetime fabric within their proximity as General Relativity addressed

Darkmatter due to its homogeneity , it renormalizes with Spacetime Geometrical deformation & curvatures across universe , so what appears to be excessive density of Darkmatter in the galactic halos , it is renormalization of darkmatter with spacetime deformation around galaxy

The lack of observable dark matter gravitational lensing in immediate proximity is due to our shared co-movement within the same local reference frame. Expecting to detect local dark matter lensing is physically analogous to an observer attempting to directly perceive the Earth's rotation from its surface; the shared non-inertial frame effectively masks the underlying relative dynamics .. hence Darkmatter is isotropic within Galactic stellar core

While The Gravitational effect of Darkmatter on Galactic arms caused by less localized reference frame correlation between galactic core & galactic arms & the difference of stellar object density between them , while Galactic arms affected by gravitational interaction of renormalized Darkmatter with Spacetime deformation around the Galaxy , causing gravitational effect on galactic arms conserve its structure around galactic core , which violates General relativity when not taken the existence of darkmatter into effect .

D) Microscopic

Quantum world Application intro

Proposed theory according to Quantum world (focusing on electrons , which represents the obvious state of quantum occurrences)

hence quantum particle due to its nature are not fully bound to spacetime constraints of locality & speed-limit

The Interpretation of the proposed theory at Microscopic Level - Role of Virtual Blackhole :

Quantum States are filled with (virtual blackholes/VBH) that electrons teleported through making electron cloud .

in case of energy quanta , energy generates more energized VBH than ground state VBH to attract electron to teleport to next level , electron teleported to higher quantum state , using newly VBH network to teleport until energy consumed

then VBHs annihilated by spatial equilibrium due to excessive number of VBHs making unstable quantum state so the excess VBHs accrete each other emitting radiation when destroyed , the exciter VBH network that allow the particle to teleport to higher quantum states annihilated by spatial equilibrium its radiation generate downward VBH attract particle to transport to its original ground state ... & downward VBH annihilated by Dark Energy expansion (These mechanism is discussed in detail in Atomic stability subsection) .

But Determining Surface Tension of Dark matter that have the same scale as in cosmic scale is impossible , hence determining distance differences between Quantum States , Orbitals & Magnetic states were geometrically Impossible , & can't rely on Quantum states in verification of scale constancy of Dark-matter surface tension

You can pinpoint the error of determining Dark-matter Surface Tension in Main Quantum states level alone

Taken Hydrogen Atom as an example

Quanta Energy = 10.2 eV = 1.63*10^-18 j

Radius of Bubble (Rb) = (n2^2 - n1^1)*Bohr Radius

= 3*5.29*10^-11 = 1.587*10^-10 m

$$\sigma = \frac{\Delta E}{4\pi R b^2}$$

= (1.63*10^-18)/(4*π*((1.587*10^-10)^2))

= 5.15 kg/sec2

Dark-matter (Dark-matter surface tension drops drastically at quantum level

& This is physically unacceptable , because Dark-matter doesn't interact with any matter not even in atomic or subatomic scale , its only observed interaction is barely through gravity upon intergalactic clustering formation

& also to confirm methodological & mathematical invalidity of determining dark-matter surface tension magnitude at microscopic scale & checking its compatibility with cosmic scale , Energy difference between Orbital n6 & n7 in same Hydrogen Atom is only ≈ 0.1 eV & Main Orbital Distance exceeds 6 angstroms (> 6*10^-10 m)

Meaning lesser values for Dark matter for the atom when excited

& that values differ from element to another due to difference in Quantum states Energies due to difference in atomic & mass number of each element … which will lead to different values of Surface Tension , which cannot be accepted physically

The reason of this error

Applying main quantum states alone neglects the order of orbital , magnetic , spin energy level order (for example Orbital 3d is preceded by Orbital 4s) which detecting them by spatial metric scales is structural error

Conclusion :

The attempt to extract a surface tension from atomic orbital separations is **conceptually invalid**, not merely inaccurate, because atomic orbitals do not possess uniquely defined geometric radii once sub-orbitals, magnetic quantum numbers, angular momentum, and saturation effects are accounted for.

The calculation presented here is intentionally shown to demonstrate the **magnitude of the conceptual error** that arises when such a method is applied.

Primary Electron Characteristics

Proposed model to solve the paradox suggests the following :

- According to Proposed Framework Principles , Electrons are considered as Point Charge (Dimensionless) . - (QFT considered Electrons Point Charge , proven , when studying Coulombs Scattering) -

1- Hence to Evade The Assumption of electrons with mass 9.109*10^-31 kg has an infinite density due to dimensionless nature (According to Coulombs Scattering Results) , given mass is the masses of infinite number of generated microscopic (Virtual Blackhole) within the Quantum state .

Duration of equilibrium spatial collapse due to accretion of such blackholes or Dark energy expansion annihilation duration is less than Planck scale (lead to zero second) , which means Blackholes existence duration are less than Planck scales (lead to zero second)

2- Infinite Virtual Blackholes (VBH) doesn't earn mass … as electrons are teleported before & after virtual confined extra universal void (VCEUV) , rotating in a loop , so energy difference equal zero all time

3- Electrons in (ground state) doesn't consume or radiate energy , hence it teleported , not rotating nor moving

4- Estimating electrons , Void , surface tension of Dark-matter quantically through main orbital energy states is not accurate nor representing actual observation & measurements , hence it didn't consider states of sub Orbital numbers into respect (n1s , n2s , n2p , n3s , n3p , n4s , n3d , n4p , n5s … etc) & these quantum states has not geometrical characteristics capability to be measured .

5- Voids generated are saturated with ground state electrons , there's no room for additional electrons to teleport .

6- when Electrons elevated from ground state to higher one comparatively bigger virtual blackholes formed from the limit of Quantum state level & next level , while comparatively small virtual blackholes formed across all levels occupied in the atom , that comparatively big virtual blackholes attract electron to be teleported to next orbital level , while Electron

teleporting in the new state , new VBHs annihilated due to spatial equilibrium collapse (as it is explained before in cosmic section) , emitting energy . some of it consumed in photon creating blackholes in the way back out of New Quantum state level returning to original ground state where original VBH were not saturated with absence of electron excited , attracting the excited electron to teleport back to its original quantum state , Dark Energy annihilates the (way-back) VBHs .

Determining duration of excitation as duration of electron being teleported to higher quantum state through frequency of photon emitted by electrons excited since applying the formula

$$\Delta E = h * f$$

Where ΔE is difference in energy between different orbital state , (f) is frequency , so

T = 1/f = h/ΔE

In case of Hydrogen Atom duration of excitation of electron from n1s to n2s

T = (6.62 * 10^-34) / (1.63*10^-18) = 4.06*10^-16 sec

Explanation:

When Electrons excited (earn quanta energy) , that energy generates more energized VBH than ground state VBH to attract electron to teleport to next level , electron teleported to higher quantum state , using newly VBH network to teleport until energy consumed

then VBHs annihilated by spatial equilibrium due to excessive number of VBHs making unstable quantum state so the excess VBHs accrete each other emitting radiation when destroyed , the exciter VBH network that allow the particle to teleport to higher quantum states annihilated by spatial equilibrium its radiation generate downward VBH attract particle to transport to its original ground state ... & downward VBH annihilated by Dark Energy expansion

7- What is claimed about electrons mass , is actually the mass of infinite virtual generated/annihilated Voids representing the teleportation path of that certain electron with its ground state orbital which clarified in formula

Mass of Electron = 9.109*10^-31 (17) $= \int_0^C \frac{dE}{c^2} = \int_0^C 8\pi\sigma c^{-2} r \ dr$

Hence E = $4\pi\sigma r^2$ (where r is (Void bubble radius)/(teleportation distance) across all Main , Orbital , magnetic level , Spin Level .

Where C is Quantum State level Circumference in meters (SI)

“The electron does not possess an intrinsic rest mass.
Its observed mass is an emergent, dynamically sustained quantity arising from continuous VBH generation and annihilation along its teleportation network.”

8- Microscopic model proposed doesn’t result on double counting effect , as electron existence in multiple location is a visionary image related to observer , exactly as in relativity , relative speed of an object varies according to the observer

& electron itself - as it is explicitly stated in the framework , is dimensionless point charge (as it is verified in Coulomb Scattering study) , so possession of mass meaning infinite density , framework does not allow , so electron in the proposed framework considered massless , & what is considered as electron mass , it is accumulative VBH mass created that electrons teleported through

Interpretation of Diffraction is applicable with applying virtual blackholes/Voids

9- the same interpretation of electron cloud .... Quantum Mechanics assume that this is waves of probabilities of electron existence , while Proposed model has different approach , as electrons appear as recurring existence due to teleportation , which is explained in the following explanation

Zero Point Energy theory when translated to the proposed framework .

10- Zero Point Energy Proposed Model Interpretation :

A) as temperature decreases near absolute value (Zero Kelvin) , Electrons in ground state emit energy , energy emissions generated extra virtual blackholes which not confined by quantum energy state , battling existing virtual blackholes for attracting electrons to teleport , electrons teleported in different paths , extra teleportation allows for surpassing energy barriers , translated in superconductivity .

& super fluidity , Zero Point Energy checked & complied

- $dE = 8\pi\sigma\Delta r\ dr$ (as Temperature decreases to near absolute value , where electrons at quantum state emitting energy to achieve Equilibrium with surrounding

- E = - $3/2 * K_b * T$ (18)

Where (K_b) is Boltzmann Constant , & (dT) is difference in temp.

Derived to the Equation

$-\frac{3}{2}K_b\ .dT = 8\pi\sigma r\ dr$

B) as External Pressure is applied , it injects mechanical energy into atomic quantum states , Orbitals & Dark-matter medium , Generating more virtual blackholes and compressing Orbitals so that virtual Voids overlap & emerge , fuses into bigger blackholes , surpassing Planck scales to measurable scale , adding teleportation distance to the electronic system , increasing conductibility .

C) as Pressure increases , Density & Connectivity of the network increases , so the temperature at which Superconductivity & Super fluidity also increases .

Where E = PV = Nmv^2(where mv^2 equivalent to Kinetic Energy) (19)

Where

PV = 3/2KbT

$-\frac{3}{2}KbdT = -d(PV) = 8\pi\sigma r\ dr$

So Pressure applied increase Temperature of Superfluidity & Conductivity

11- Quantum State Energy = The Sum of Potential Energy of (Virtual Blackhole/Void) + The Sum of Kinetic Energy of Spin of the Virtual Blackholes + magnetic field of the electron forces it to take certain variable sets of Void paths due to repulsion with other electrons = Constant

12- Virtual Void/Blackholes spinning caused by spinning of Electrons

Observer Effect

13- Wave Nature of Electrons such as deviation/superposition is a result of infinite virtual blackholes causing infinite teleportation that would make electron appear in multiple coordinates at the same time .

The Observer or detector effect can't occur without energy interference , energy will increase VBH transform it from virtual to near actual (closer to Planck scale) , causing teleportation distance to increase dramatically , fusing other virtual blackholes , which resulted in detecting electron as single particle , which would be translated as (collapse of electrons wave function) Proposed Theory would dig deep into detail , interpreting The new version of Double Slit Experiment (Delayed Choice Quantum Eraser) & What seems to be future determine state of the past , after explaining the following Entanglement Subsection in the Proposed Model point of view.

Entanglement

14- Entangled Particles are particles that their quantum states are potentially correlated , such as their angular momentum where conserved , as two entangled particle super-positioned correlated that both of them teleported through same VBH together affecting their spin to be opposite of each other , so their angular momentum conserved

This entanglement correlation lasts wherever being apart from each other .

But their correlation could be violated hence one of them entering similar spinning blackhole .

& which happened also in detection state

that comply and validate & interpret Bell's Inequality Test (BIT)

& that also comply to QM standard as there's no capability - according to our understanding of physics – send information through entanglement in controllable state .

if two electrons teleported through same VBH at the same time , they would share same quantum state

& hence they were point charged , same as virtual blackholes (Theoretically)

Pauli's Exclusion Principle are being violated (requiring that virtual blackhole generated exactly where both electrons will be (in Diffraction or Interference Phase) , this condition generates blackholes within electrons proximity themselves as teleportation distance is virtual too (Beyond Planck scale) … these blackholes form a bubble , that bubble has different characteristics

Let's call these kind of VBH as (Stucks)

These (Stucks) have a bubble sphere with surface tension , but this surface tension differs entirely from Cosmic , or even Virtual Black-holes .

Their medium isn't Dark-matter anymore , their medium is (electrons) themselves

Because electrons are massless (by their own) & dimensionless

So their density & surface tension are indeterminate

Their Stucks Void bubble surface tensions = (indeterminate)
so it could be any value , depending on distance between entangled couples , as electrons in this case has the upper hand on Stucks

So Information could be send at any distance , even if they were halfway around the universe .

This is how Entanglement is being generated or created

Because both particles were entangled , they have correlated quantum states , that information couldn't be decoded by any third party , Only their entangled partner is the decoder , even if it forming Stuck Bubble Sphere , this information has no use but for entangled partner

& that information may be sent through electron infinite spin speed , as it transmitted through stuck bubble sphere circumference infinite times … & only entangled partner which has another stuck , catch it & decoded .

But , There is only one hypothetical way that could make sending information through entanglement in controllable state a possibility , depending on one condition , According to the proposed model

In order to send information & preserve it in controllable conditions , both electrons should be operated at very low temperature (near zero Kelvin) , emitted energy from electrons generated additional non-spinning VBH … which even when they interact with entangled electrons , they won't affect their spin

So (Theoretically) sending information entanglement preserving it in controllable condition is a possibility

That possibility increases as temperature decreases near (0) Kelvin more & more … hence more non-spinning VBH are generated from electrons emissions .

That was derived from Wien formula (20)

T*$\lambda = 0.0029$

As

T=0

$$\lambda = \frac{0.0029}{0} = \infty$$

As system energy = h/ λ = h/∞ = zero

Atoms Quantum States emit energy (As of Hydrogen for example)

Quantum State Energy = 10.2eV = 1.63422*10^-18 joules

Its wavelength $\lambda = \frac{hc}{E} = 6.62607015 \times 10^{-34} * 299792458/(1.63422 * 10^{-18})$

= 1.2155 * 10^-7 m

That is the amount of energy that Quantum State emit in a system (Theoretically) reach 0 Kelvin to balance the surrounding of reaching (Zero Kelvin) , Which is equivalent to the amount of energy that the electron is stabilized in its quantum state , where occurrences such as (Superfluidity , Superconductivity , Entanglement) reach the efficiency of 100% , so emitted energy creating extra (VBH) Equivalent to stable quantum state of Hydrogen Atom without excitation

As Entanglement Control Parameter Efficiency (η) = 1

Computing the temperature where entanglement has (Zero Efficiency – Uncontrollable entanglement)

Where T*$\lambda = 0.0029$

T = 0.0029/ λ

= 0.0029/(1.2155 * 10^-7) = 23,858 Kelvin

(η) = 0

But the Efficiency Curve isn't linear

According to Friedmann Equation , Radiation ratio is (0.00009) & parameter of ((a)^4)

Radiation parameter is included hence temperature is main contributor of control efficiency

(η) = 0 T = 23,858 Radiation ratio = 0.00009 (Rad parameter) = ((a)^4)

(η) = 1 T = 0 Radiation ratio = 0.00009 (Rad parameter) = ((a)^4)

Where X = (η) , U (is Constant)

$$T(\eta) = \frac{U}{(0.00009a^4)^n}\left[\frac{e^{-\frac{\eta}{(0.00009a^4)^n}} - e^{-\frac{1}{(0.00009a^4)^n}}}{1 - e^{-\frac{1}{(0.00009a^4)^n}}}\right]$$

As (n) is element electron quantum states occupied

& This is minimum precision of computation of a formula to describe Entanglement control efficiency that represents the real world testing & observations

Delayed Choice Quantum Eraser Experiment :

15- The Recent Version of Double Split Experiment (Which is also called **(Delayed Choice Quantum Eraser)**) confirmed that detection of the particle will collapsing its wave function even after its being recorded , & that will determine the behaviour of the recorded state whether it is wave interference or double slots reflecting the double slits (as it behave as particle) ----------(21)

The Proposed Framework interprets that as previously mentioned as energy interference as explained earlier in Observer Effect Subsection
& due to Entanglement , the detected particle will transmit the data to the recorded entangled partner to correlate its quantum state to its counterpart , so the wave state collapses as Standard Model confirms (Or their VBHs combine to get near Planck limits to near measurable scales as The Proposed Model interprets) , & preserve its particle state (Or their teleported distance being near measurable as Proposed Model interprets) , even after one of entangled partners being recorded (meaning after it is being received) .
& Because Transmitting information is faster than light (instant/teleport) so it appeared to the Observer that Future can determine the past in The new version of Double Slit Experiment (The Delayed Choice Quantum Eraser).

Nucleus Application

16- Applying Microscopic effect from electrons to nucleus

sum of VBH responsible for quark teleportations across nucleons proximity give the mass of the quarks , that masses are larger than that of electrons due to limited & narrow proximity , which allow VBH to fuses together making larger one & further teleportation distance , But the narrow confinement restricted it , so the energy measured of quarks is near 100 times of electrons , no problem with that assumption .

The Reason of Narrow Proximity
(Stucks) are being formed due to generation of Virtual blackhole - in this time - on three quarks
This will generate Stucks inside every quark which they share quantum states upon generation , that is the foundation of what it is commonly used to be called (Gluons) .

(Gluons) are a three way entanglement processes , three quarks quantum states correlated with each other .

1st (Three-Way-Entangled-Quark) sends information by its stuck to take opposite quantum state (Up to Down quark)

But hence there are three entangled partners , that will lead to continuous endless conversion .
Stuck teleported information from 1st Quark who's (down quark) convert 2nd quark to (up quark) , 2nd stuck teleported information convert 3rd quark to (down quark) , 3rd stuck teleported information convert 1st quark to (up quark) , 1st stuck teleported info convert 2nd Quark to (down quark)

(3 way entanglement (gluon) change quantum states of quarks from Up to Down instantly) , that continuous charging fluctuations generate the mass of nucleons .

This model actually aligns perfectly with standard model ... that means when neutrons are converted to protons , protons converted to neutrons & vises versa .

It's simplest form of entanglement to form single nucleon …. But in Nucleus…. The Generated Virtual Blackhole interact with all nucleus quarks , so signaling opposition & variation does not only confine single nucleon , but also confine all nucleus nucleons together

the signal change due to quark (signal & quantum) change , teleportation path change & angular momentum change of VBH due to quarks continuous conversion , it all leads the whole package (nucleon) to oscillate & have a kinetic energy , of Intrinsic Compton frequency of 2.27*10^23 Hz

(Being Determined Out of Rest mass energy equilibrium mc^2=hf)

Where

m is Mass of nucleon in kg = 1.67492749804*10^-27 kg

c = speed of light = 299792458 m/sec

h = Planck Constant 6.62607015*10^-34 j.sec

(Strong Force time scale of 4.4*10^-24 sec)

& Wavelength calculated of $(mc^2) = \frac{h^2}{16\pi^2 m.\lambda^2}$

where

$$\lambda = (\lambda^2)^{0.5} = \left(\frac{(h^2)}{16\,\pi^2 m^2 c^2}\right)^{0.5}$$

$$= \left(\frac{(6.62607015 * 10^{-34})^2}{\left(16(\pi^2) * ((\,1.67492749804\ * 10^{-27})^2) * ((299792458)^2)\right)}\right)^{0.5}$$

= 1.0500971*10^-16 m

That's exact half of the assumed Compton wavelength

$\frac{h}{2\pi mc}$ = 2.10*10^-16 m

For Proton $= \left(\frac{(6.62607015*10^{-34})^2}{\left(16(\pi^2)*((\,1.67262192595*10^{-27})^2)*((299792458)^2)\right)}\right)^{0.5}$

= 1.05154455026*10^-16 m

while experimental measured charging radius of proton = 8.409*10^-16 m (8 times nucleon calculated wavelength)

while Charging radius doesn’t refer to nucleons wavelength itself
The Geometric Summation

Experimental measurement of the proton's charging radius does not measure a single linear path. It measures the "Charge Cloud"—the probability summation of the charge's location in 3-dimensional Cartesian space.

To find the effective radius of this cloud , that must account for the oscillation extending through all spatial degrees of freedom. In a 3-dimensional system (x, y, z), the volume expansion of the oscillation corresponds to the summation of the 8 spatial octants (2^3 = 8).

R(predicted) = 8*wavelengths calculated

Consider our Nucleon as a small cube surrounded by another 8 cubes in Rubik’s Cube ... Scattering measure the sum of 8 wavelengths surrounding the nucleon under study .

& that align with spheres , as central sphere separated by 8 spatial Octants in Cartesian Coordinates

& there’s no interference or overlaps of wavelength or we violate PEP (Pauli’s Exclusion Principle)

3. The Result & Verification

Substituting the calculated wavelength:

R(predicted) - neutron = $8 \ * \ 1.0500971 * 10^{-16}\, m \ = \ 8.4007768 \ * 10^{-16}\, m$

R(predicted) - proton = = $8 \ * \ 1.05154455026 * 10^{-16}\, m \ = \ 8.41235640208 \ * 10^{-16}\, m$

Comparison with Experimental Data:

Recent high-precision experiments (muonic hydrogen spectroscopy and CODATA 2018) have revised the standard proton radius.

- **Experimental Value (CODATA 2018):** CR = **0.8409** femtometer (22)
- **Predicted Value (Proposed Framework):** CR = 0.8401 femtometer
- **Predicted Value (for Proton):** = 0.8412 femtometer

Conclusion:

The framework predicts the proton charge radius with 99.9% accuracy. This suggests that the observed size of the proton is simply the 3-dimensional geometric summation of its internal oscillation wavelength (99.9%) is due to proposed model approximation .

17- Interpretation of 8-Octants is valid , it represents 8 points of contact at rest state , argument of spherical points of contacts increases to 12 point of contacts is irrelevant , hence that geometrical state occurred only under compression , so wavelength decreases , & that is not the subject of the discussion .

18- The Continuous conversion of Proton & Neutrons , forces electrons teleportation paths in its orbital to follow protons conversion .

Decay

19- Proposed Theory is inclined perfectly with QM , as Stable Nucleons should have least generated (potential energy/mass) out of Generated VBH & all forms of Entanglements , if it's not , decay will occur the same way as QM states , to elements byproducts that will represent least potential energy possible & equivalent spin vector (in case of Beta decay for example) , which QM gives perfect rules (such as Fermi's Golden Rule) , which it has no contradictions with proposed theory .

Magnetism

20 - Proposed Model doesn't contradict with magnetism , As Magnetism Properties relies on Unpaired electrons tended to conserve its angular momentum , & system tendency to consume minimum energy as explained earlier at decay discussion interprets heavy elements lacking the fill of more than single orbitals , hence resultant vector of entanglement & angular momentum (which is described by the current standard model as (charge) , where in same spin direction that lead to increase in wave length in between (same charged generated virtual photons resulting in repulsion by Standard model interpretation) .

So heavy atoms consume minimum energy as electrons were occupy different orbitals without filling them one by one (particularly at high orbital levels) , so minimum energy applied - PEP (Pauli's Exclusion Principle) being conserved … so heavy elements are tendent to have high magnetism characteristics .

Planck dimensions Limit , Photons & Charge Interpretation

21- Proposed Framework interprets earlier that Electromagnetic wave (Light) is The Cosmic Capillary Wave at Resonance state

Photons is Quantization of Electromagnetic Wave , as EM action cannot be decreased down the Planck limit of 6.62607015*10^-34 j.s , which equals photon , a quanta EM .

Photons are analogous to electrons & quarks , as they are affected by VBHs , Teleport , behave the same way

Since at Planck dimension limit it is not possible to distinguish between Particles & Waves , Existence & Nothingness

& That forces our Proposed model to reinterpret Planck limits according to its proposed phenomenon

22- Planck limits is The proximity of VBHs occurrences that affecting particles & waves to conserve its recurring occurrences & appearances (as wave state according to standard model) .

Reaching dimension Planck limits would convert VBHs to near measurable & massier VBHs , Teleportation distances increases , recurring occurrence cannot be determined due to increase of teleportation distance (as Standard model state , that it will preserve the particle state)

23- Proposed Model interpret Electric Charge of any (particle) as the resultant vector of spin vector & entanglement vector & That will be discussed by examples in the later section of Microscopic Conclusion

Atomic Stability

Accretion Limit

24- Accretion limit reach (of $R_{B/CEUV} = R_{Schwarschild}$ annihilates Virtual Void bubble/Virtual CEUV near instant time (at Planck time) as it does in cosmic Void formation in measurable time , this occur in upward excitation VBH , forming downward VBH & in VBHs created in excitation quantum states that radiate due to reaching accretion limit spatial equilibrium .

System stability maintained .

This occurred at Quantum Jumps between Quantum energy states & VBH in higher quantum state in excitation

Dark Energy Contraction

25- Annihilation through contraction (Due to Dark Energy expansion rate) :

Because (VBH) where beyond Planck limits , VBHs where vanished & generated instantly , this occur when the particle return back in its ground state , & when particle return back to its ground state , VBHs created from radiation of (upward-generated VBHs) when annihilates due to reaching spatial equilibrium by accretion limit

Energy Quanta

26- if the atom earn energy quanta , it generates 3 kinds of VBHs

High Density Upward VBH chain

More massive VBH chain that accrete particle from its ground state to the excited state

They annihilated through Accretion limit spatial equilibrium due to its high density of VBHs , emitting photons generating (Low density Downward VBH Chain)

Quanta VBHs

Extra VBHs generated in the excited Quantum state for particle to teleport , they annihilates due to high density since excited quantum states overcrowded by its ground state VBHs & excited state VBHs , annihilated by Accretion Limit Spatial Equilibrium

Low Density Downward VBH chain

Low Density VBH chain created out of photons emitted from annihilation of Upward VBH chain

Its density is low since after quanta energy consumption , the higher quantum state is not excited anymore , meant all of its VBHs are saturated , excited particle being accreted back to

its ground state , Downward VBHs annihilated by dark energy local expansion emitting last photons

Quanta Energy is conserved

Quantum States & Probabilistic Nature

27- According to The Proposed Model

Quantum Energy States forms & defines Quantum Particles Wave function inside the atom

As Quantum states , Orbitals defines particles wave function .

As Wave Function is defined as VBHs Teleportation routes inside the atom
& magnetic number represents the vector magnitude between entanglement vector & spin vector of the particle which affect increasing & decreasing of wavelength between particles , which will further direct & redistribute teleportation routes of generated VBHs , as it will be clarified & discussed further in Conclusions sections .

28- According to The Proposed Model

Probabilistic nature of particles is generated out of dimensionless particles & intrinsic generated VBHs that operate beyond Planck limit , which will cause Impossibility of determination of coordinates & momentum at the same time , & that explains Quantum fields operate perfectly within its domain

As any dimensions beyond Planck length , Planck time (Planck Spacetime) will lead to zero , so it cannot be determined , but its effects (generated mass & energy) can be measured

Bosons

Definition

W&Z Bosons

29- These Bosons are The Quantum action (Energy Generated in time) applied to generate/dissimilate Quantum Entanglement of particles , whether it is W,Z

Higgs Bosons

30- Higgs Boson is The Quantum action (Energy Generated in time) applied for Quantum state to generate VBHs Network for each particle , that gives particles their intrinsic mass

31- Its decay byproducts distribute Quantum states energy & proximity

Life Span

32- These Quantum action occurs due to quantum transitions , that explains their extremely short lifespan

VIII) Conclusion :

1 - Cosmic

A - Speed of Light Constancy Nature

Speed of light (c) is not a constant by default ... it is a (wave) its speed dependent on the density & surface tension of the containing medium @ percussion wavelength

So , to be Generalized relativisticly

$$\forall = \sqrt{\frac{2\pi\sigma}{\lambda_{l_{cop}} * \rho}} \ ,$$

as $\forall = \ is\ Ultimate\ Wave\ velocity\ of\ a\ medium\ not\ be\ dispersed\ or\ surpassed$

$\lambda_{l_{cop}}$
$= \ is\ The\ Wave\ length\ that\ apply\ reasonance\ without\ Collapse\ which\ being\ applied\ @\ COP\ (Center\ of\ Percussion)$.

So in Any Object moving in any Medium , there general relativistic characteristic should be displayed this way

Such mass $M \ = \frac{m}{(1-\left(\frac{v^2}{\forall^2}\right)^{0.5}}$ Which is considered a generalization of Lorentz Invariance in different universes or environments (or mediums) (23)

Also time & length , etcetera .

B - Centre of Percussion :

The wavelength of the universe is not random, & it is governed by the "centre of percussion" of a uniform rod or ideal physical pendulum, which is mathematically located at two-thirds of the length 2/3 L

Physical and Mathematical Precision

In classical mechanics, for a body (or medium) to be in perfect resonance with zero reaction force at the fulcrum, the point of application must be at the centre of the shock.

For a uniform rod rotating about its end, the moment of inertia is

I = 1/3 ml^2 (24)

The distance to the centre of mass is ½ L

The centre of impact (L-CoP) is calculated using the equation:

L-Cop = (1/3 mL^2)/(1/2 mL) = 2/3 L

Applying that to universe , the perfect resonance it makes the system doesn't have any collapse rebound , it's fulcrum at 2/3 of the radius as Center of Percussion of a Uniform Cosmic Medium

And Choosing Point of Percussion as solid rod behavior , is a logical & excellent choice of studying medium that has surface tension of 2.994*10^16 Kg/sec2 (Noting kg/sec2 & J/m2 is equivalent units of course) , & that indicates dual characteristics of the universe of hydrodynamic nature with solid properties (such as astronomical amount of surface tension)

Also there would be an argument that this formula only applied for solid rod , but the Universe for the proposed model assumption act as solid sphere , & for solid sphere , CoP would be at 7/5 R , not 2/3 R

But The Proposed Model doesn't study a homogeneous solid sphere … it studies a sphere with solid surface tension of 2.994*10^16 Kg/sec2 (J/m2) & near zero density of $8.5383542 * 10^{-27}$ kg/m3 magnitude

So the study should be about (solid spherical (shell)) , not solid sphere

$$I_{sphereical\ shell} = \frac{2}{3} * mR^2$$

$$L_{CoP} = \frac{I_{Spherical\ Shell}}{m * R} = \frac{2/3 * m * R^2}{m * R} = \frac{2}{3}R$$

Applying that will result in Universal matter percentile compatible perfectly with what's actually observed experimentally of ~ 0.3183

C - Universal Surface Tension & Critical Density are not a real constants

they were Our (lifetime) constants , as Universe expand , Surface Tension & Density both were inversely proportional to ΔDc^2 & ΔDc^3 respectively & also percussion wavelength increases , so It's Capillary wave speed (c) of our Universal Matter remained constant .

D - Misconception of Void Bubble Violation with General Relativity & Observations

to address misconception & refuting it very clearly , Void/Void Bubble operating out of spacetime/Universe completely Generated Void Bubble from Blackhole formation doesn't affect any stellar object within Blackhole Void Bubble range or interact with it neither chemically nor gravitationally , it doesn't interfere with gravitational lensing … Bubble Void doesn't interact with anything Outside its gate (Blackhole Schwarzschild event horizon itself … Void Bubble is The same as Void (nothingness) surrounding Our Universe … Void Bubbles is the same void , But that void is confined by Universal Dark-matter .

E - Spaghettification Effect Refutation

According to the proposed model introduced , there will be no (Spaghettification Effect) on the teleported Object through blackhole , since Blackhole Definition in Proposed framework is nothing but a topological shortcut , & also there's no damage will occur to the teleported objects outside the Void bubble/CEUV .

F - Relativity Reservation & Speed of light constancy origination

Out of Capillary wave Absolute Velocity formula

$$c = \sqrt{\frac{2\pi\sigma_{UM}}{\lambda\rho}} \qquad \boldsymbol{Where\ \sigma_{UM} = \frac{E_{UM}}{A}}$$

Where $\lambda = \frac{2}{3}D_c$

Where D_c $is\ Universal\ Comoving\ Distance$

$$c = \sqrt{\frac{2\pi * (um) * \frac{4}{3}\pi {D_c}^3 \rho * c^2 / 4\pi {D_c}^2}{\frac{2}{3}D_c * \rho}}$$

That means that Speed of light limit depended on Potential Energy of The Substrate Medium Alone (Which is in our case Cosmic (Universal) matter , which is the combination of Baryons & Dark Matter .

The constancy of Absolute Capillary wave speed is conserved , as Universal local expansion decreasing surface tension & critical density by the same rate (as critical density itself is a main factor in surface tension magnitude , as equation above explains

G - Dark matter primordial inhomogeneity/current homogeneity Assumptions

Proposed Model constitute that the source of Darkmatter is the residual energy of Primordial Quantum Fluctuations , that inhomogeneity distributions due to Big Bang Inflation dictates the newly formed Intergalactic clusters scaffolding , While Thermodynamic Entropy contradicts inhomogeneous distribution of Energy , Intergalactic clusters forms to give Darkmatter the required resilience to gain its homogeneity later phase which comply to Thermodynamic Entropy .

2 - Microscopic

A - Quarks and electrons are the same dimensionless, massless particles, and even have the same charges.

a) The energy generated by the Big Bang confined the point-particle as a quark in the nucleon and as an electron in the atom.

b) what makes it appear as if the quark's charge is taken values of one-thirds & negative two-thirds that of the electron , in reality, the one third negative charge to two-thirds positive charge is a result of the triple bonding, which causes the charge to decrease by one-third (as a resultant) combine that with direction of quark spin ... that is, as if the triple bonding occurs at a 120° angle … & being up & down quark affect the direction of spin relating to entanglement vector , that would make the charge oscillate from -1/3e to 2/3e (-60° to 120° , which the difference is 180° (difference between up & down) .

This explains the precise stability of the hydrogen nucleus, which contains only one proton.

Incidentally, the sign of the electric charge is most likely just the vector of the instantaneous entanglement between particles … (so that why electron has its negativity charge full as entanglement at 180° … & it's negativity could be resulted out of their entanglement vector as opposite to its spin vector

c) So as electrons , their charge negativity is properties of their entanglement at 180° , data teleported across entangled in opposite vector to the spin vector (maintaining Inertial Stability) , electrons entanglement are in the opposite vector of quarks triple entanglement , that is the cause of what's being described as charge contradictions between positive & negative in quantum level

& the electrons teleportation orbit around nucleus is interpretable as unifying GR with QM , this is spacetime fabric bent for nucleus , electrons attracted orbited , VBH formed to make electrons teleported around .

d) about electromagnetic attraction & repulsion … it depends on energy effect across time (Action) , as action lead to wavelength to increase , repulsion occur , if it leads to wavelength to decrease , attraction occur

so attraction occur when two particles their entanglement & spin resultant vectors are opposite to each other , the opposite vectors generating action reduces medium wavelength in between , attraction occur …. Inversely occurrence in repulsion

so

"The framework hints at a deeper unification where quarks and electrons may be the same topological void entity, confined differently. The fractional charges of quarks (-1/3, 2/3) could be interpreted as geometric vector resultants arising from a stable 120-degree entanglement triangulation within the nucleon & quark spin vector , rather than intrinsic properties. Furthermore, electric charge polarity (+/-) implies the directional vector of the teleportation path through the spacetime medium

e) while the formation (or classification) of quarks & electrons depended on energy generated after the big bang

first , Universal size resulted in energy density & temperature were enough to form triple-entangled quarks then nucleons (the way described earlier in quantum model) , after expansion & cooling , atomic electrons & orbitals were formed .

all these were gravitational formation , & gravity itself is interaction of energy resonant capillary wave speed (EM) in our universe , & EM is the interaction of surface tension & critical density & percussion wave length of the containing medium , & surface tension - density are interaction of energy & medium dimensions …. Etc.

B - Neutron/Hadron Stars formation

Standard Model vs. Proposed Model:

According to Standard Model , Neutron Star formed out of Neutrons as Gravity violate Stellar Object electron's PEP , it forces it to fuse with Protons convert it to Neutrons , as The stellar Object reaches Chandrasekhar mass limit for Iron core star described in equation

$$M_{\text{Ch}} = \frac{\omega_3^0\sqrt{3\pi}}{2}\left(\frac{\hbar c}{G}\right)^{3/2}\frac{1}{(\mu_e m_H)^2}$$

Where

$$\omega_3^0 \text{ is Lane Emden Polytrope Constant} = 2.018236$$

$$\mu_e \text{ is Atomic Weight/Atomic Number ratio (for Iron)} = 2.15384615385$$

$$\mu_e \text{ is Atomic Weight/Atomic Number ratio (for Oxygen)} = 1.999925$$

$$m_H \text{ is Mass of Hydrogen Atom (Mass of Nucleon)} = 1.67 * 10^{-27}\text{kg}$$

$$\frac{\hbar c}{G} \text{ is Squared Planck Mass of magnitude of } (2.176434 \times 10^{-8}\text{kg})^2$$

M(CH) Iron = 2.46860478509*10^30 kg = 1.24050491713 Solar Mass

M(CH) Oxygen = 2.86321302511*10^30 kg = 1.43990757975 Solar mass

According to The Proposed model , The title (Neutron Star) is far from accurate , as Gravity violate electron's PEP , it fuses with Protons in Nucleus , & Due to Gravity it will form rare 4-way entanglement of quark Hadrons (excited tetra-quark mesons) of 4 quarks of 90° angle , so it will form perfect symmetrical positive & negative charge , which lead to neutral condition

According to Proposed Model

In order to for electron to interact with proton spontaneously, it must yield & to be subjected to Fermi's Golden rule , seeking lowest mass result

This newly formed Meson is known to be lightest tetra quark meson , known as Sigma Meson of mass = 0.867 * 10^-27 kg. (25)

So , Neutron Stars , contain both Tetra-Quark mesons (which is post-proto-electro-fused particles) & Tri-Quark neutrons , which causes momentum imbalance , that will generate high momentum neutron star spin generating pulses observed

So this star should be called Hadron Star , Not a Neutron Star

1 - Where Chandrasekhar mass limit of Iron Core Hadron star according to the presented model

$$M_{\text{Ch}} = \frac{\omega_3^0\sqrt{3\pi}}{2}\left(0.5\sqrt{\frac{\hbar c}{G}}\right)^3 \frac{1}{(\mu_e m_H)^2}$$

Where

$$\mu_e \text{ for Iron at this case} = \frac{\left(\frac{0.867}{1.67} * 26\right) + 30}{26 * 1.67/0.867} = 0.86856155032$$

Where 26 is the number of interacting Protons with Electrons

& $0.5\sqrt{\frac{\hbar c}{G}} is\ Corrected\ Planck\ mass\ according\ to\ Proposed\ Model$

M(CH) = 1.8975354259*10^30 kg = 0.9535353899 Solar Mass

This is the (Theoretical) Minimum Mass required to form (Neutron/Hadron) star

Actual mass required will be discussed & derivated in the following few subsections

During the formation of (Hadron Star) …. Emission of Gamma rays occur , as represents the difference between Both sides of Interaction equation

in Hadron Star limit according to Proposed Model

$$\frac{\left(\frac{0.867}{1.67} * 26\right) + 30}{56} = \frac{1.8975354259 * 10^{30}}{Total\ Equivalent\ Mass}$$

Total Equivalent Mass = 2.4429051*10^30 kg
Mass lost = 2.4429051*10^30 - 1.8975354259*10^30 = 5.4536968*10^29 kg

Energy Emitted = 5.4536968*10^29 * 299792458^2 = 4.9015382 * 10^46 joules

This is Energy of a Supernova magnitude , which it will left behind either Neutron/Hadron Star , or a Blackhole .

2 - Where Chandrasekhar mass limit of Oxygen Core Hadron star according to the presented model

$$M_{\text{Ch}} = \frac{\omega_3^0\sqrt{3\pi}}{2}\left(0.5\sqrt{\frac{\hbar c}{G}}\right)^3 \frac{1}{(\mu_e m_H)^2}$$

Where

$$\mu_e \text{ for Oxygen at this case} = \frac{\left(\frac{0.867}{1.67} * 8\right) + 8}{8 * 1.67/0.867} = 0.78869052314$$

Where 8 is the number of interacting Protons with Electrons

& $0.5\sqrt{\frac{\hbar c}{G}}$ $is\ Corrected\ Planck\ mass\ according\ to\ Proposed\ Model$

M(CH) = 2.30132443046*10^30 kg = 1.15733424716 Solar Mass

This is the (Theoretical) Minimum Mass required to form (Neutron/Hadron) star

Actual mass required will be discussed & derivated in the following few subsections

Sigma Mesons existence in Hadron Star is arguable , hence it is unstable particles that will lead to quick decay of neutron star , but due to excessive gravity , formation & decay of sigma mesons occur simultaneously to achieve sort of equilibrium in Hadron star state , evading quick decay .

Sigma Meson Creation - Solving Paradox

Sigma Meson Which is designated as (f0(500)) - According to QCD - has a mass of 0.867*10^-27 kg ≈ 486.35 MeV

Sigma Meson is a subatomic particle contain 4 quarks & according SM , it contains the following :

1-Upquark

2-Up Anti quark

3-Down Quark

4-Down Anti quark

The Sigma Meson is less in mass than Baryons , although it contains one more quark

Proposed Model suggests the following reasons :

A -Sigma Meson constituents interaction occupies larger proximity , causes less volume of VBHs generated .

B - Vector Geometry play role in Interactions between constituents quarks

C - Quadratic Entanglement (That substitutes Gluons) is far less in frequency & generates less Energy than Triple Entanglement (Baryon Gluons) , which forces unstoppable conversions of three quarks .

Neutron/Hadron Star Decay constituents:

Neutron Side of a Hadron Star (as proposed model calls) will decay into Stable Proton + Electron + Electron Antineutrino

Sigma Meson Side will decay into 2 states Pions : (26)

1 - Positive & Negative Pions of 66.67% occurrence

2 - Neutral Pions decay results , probability of 33.3% occurrence

First State will decay into Muon & Anti Muon with corresponding Muon Neutrino/Anti Muon Neutrino Counterparts 99.8% occurrence

Or decay into electron/positron with corresponding electron neutrino/anti neutrino counterparts 0.2% occurrence

Second State will decay into Gamma rays 98.8% occurrence

Or in electron + positron + gamma rays 1.2% occurrence

This constitutes in the end of the final constituents

Protons + electrons + Gamma rays

This constituents will form Hydrogen Dust under gravity , will form smaller stars which will end in white dwarfs at the end of its life , Such as The Sun predicted destiny as a White Dwarf.

Neutron/Hadron Star Lower & Actual Upper Mass Limit

1) Lower Mass Limit

As it is being studied & discussed in the framework

The Lower Mass limit of Hadron/Neutron Star Formation = 1.8975354259*10^30 kg = 0.9535353899 Solar Mass

This Mass magnitude is considered The theoretical maximum limit of White dwarf formation & minimum limit of Hadron/Neutron Star formation

But this mass is only enough if Sigma Mesons is formed & produced automatically - off hand -

But Actually it needs more energy to dismantle strong interaction (3 Way Entanglement binding force) to lesser interaction (4 Way Entanglement)

While , Ratio of The Mixture to Atomic mass number for Iron = $\frac{(0.867*26) + (1.67*30)}{55.845*1.67} = 0.77890954593$

The minimum limit of forming Hadron (Iron) Star will be = (1.8975354259*10^30)/ 0.77890954593

= 2.4361435*10^30 kg = 1.22513465126 Solar Mass

While , Ratio of The Mixture to Atomic mass number for Oxygen = $\frac{(0.867*8) + (1.67*8)}{15.9994*1.67} = 0.75960932367$

The minimum limit of forming Hadron (Oxygen) Star will be = (2.30132443046*10^30)/ 0.75960932367

= 3.0296158*10^30 kg = 1.52359140445 Solar Mass

Calculating 2 Lower limits of Neutron Star formation will be discussed independently in the following next few subsections .

2) Maximum Mass Limit

The Maximum Hadron star mass follow the same principle

While The Maximum mass limit of Hadron/Neutron Star Formation = The minimum mass limit of Blackhole formation

$$n = \left(\frac{\left((6.62607015*10^{-34})*(299792458)\right)}{\left(2*(\pi)*(6.6743*10^{-11})*((1.67492749804*10^{-27})^2)\right)}\right)^{\frac{3}{2}} = 2.1940608727 * 10^{57} neutrons$$

Where for the Sun

No of Solar Neutrons = Solar Mass/ Neutron Mass

$= 1.98847 * 10^{30} / 1.67492749804 * 10^{-27} = 1.18719765621 * 10^{57}$ neutrons

Theoretical Blackhole mass limit = 2.1940608727/1.18719765621 = 1.848100744828204785123057044785 9 Solar Mass

(This is The Theoretical Limit of Blackhole formation as it is presented in Studying Blackhole Subsection

But this value depended that Blackhole is formed due to Gravitational effect on (Neutron) Star

But The Proposed Model deduce the violation of PEP of electrons forming (Hadron) Star .

Mixture of Neutrons & Sigma Mesons according to the presented Model of Iron core

Ratio of The Mixture to Atomic mass number for Iron = $\frac{(0.867*26)+(1.67*30)}{55.845*1.67} = 0.77890954593$

The Actual Blackhole Formation minimum mass limit

= Theoretical Blackhole formation minimum mass limit of Baryon / Ratio of Hadron Star atomic mass mixture to Iron Atomic mass

$= 1.848100744828204785123057044785 9 / 0.77890954593$

= 2.372676974476689971567539811350 1 Solar Mass

So , minimum mass required to form Cosmic Blackhole **= 2.37267697447 Solar Mass**

This excess in Mass Magnitude , As Formation of Blackhole counteract Weak Interaction , not Strong Interaction as Standard Model Presents

Important Note :

This maximum limit occur only for Iron , this maximum limit is also minimum limit of blackhole formation … this limit won't be achieved by Oxygen composition , since under gravity oxygen will fuse into iron , stabilizes gravitational energy with nuclear fusion

C - Elementary Particles Unitarity

1 - Neutral Elementaries

All Previous iterations explain the occurrence of neutral particles such as neutrinos , which is the same as electrons & quarks as same particle phenomenon , but their entanglement vector & spin vector are 0° , Anti Neutrino its spin-entanglement vector angle where also 0° but in Opposite direction (As Negative Zero hypothetically) .

2 - Elementary Particles Generations

that iterations also explains next generation particles of All kinds charm-strange , top-bottom fermion quarks , also muon-tau electrons & neutrinos were all also follows same principles & rules in proposed model , as these generations are recognized as (VBHs) attracted each others that resulted in attracting other (VBHs) which originally were outside of their proximity (so called next generations particle) were just same particle but uses more VBHs to teleport (That interpreted their increase of mass) , as VBHs contacted each other , they became real (surpassing Planck scale) , so Wave Function (Actually Teleportation Distribution Paths in Precise iteration) collapses (actually particle teleportation distance got out of reach , as VBHs became close to RBHs (Real Blackholes) , & hence Reaching Accretion Limit virtually in quantum level annihilate their Void Bubble in immeasurable rate (beyond Planck Limit) , so these generations annihilated immediately .

3 - VBH Physical entities Ontology

virtual black holes are not independent physical entities, but transient mechanical topological configurations of the underlying medium of Darkmatter. They possess no standalone observables, but manifest only through their collective macroscopic effects.

4 - Blackhole Formation & Conserving information

Conserving Information is applied in Proposed model as suggest Blackhole as a topological shortcut .

3 - General

A - The framework does not rely on renormalization in the quantum-field-theoretic sense .
Instead, scale dependence arises from changes in geometric configuration and state saturation of the same underlying medium

& Proposed Theory uses Geometric Scaling already as it is represented explicitly

B - Undefined speed ≠ Infinite speed /Speed ≠ Distance
In a pure vacuum (no medium to contain) = EUV , the absolute speed of the capillary wave is undefined.

In a vacuum constrained by a medium (such as the vacuums resulting from the collapse of stars into black holes) = CEUV , the absolute speed of the capillary wave is infinite.

During entanglement, the point vacuum responsible for information transfer (Stuck) has an undefined distance for instantaneous teleportation. This is unrelated to whether or not there is an infinite speed (since an infinite speed is achieved simply by confining the vacuum within a material space - in space condition it will be darkmatter , in entanglement it will be electron/quark matter-). This is because the electron and quark (in the presented model) have zero mass and therefore zero potential energy, and their surface tension is undefined because their mass, potential energy, and dimensions are all = zero.

So Absolute Capillary Wave Speed in Void Outside The Universe = $\frac{\sigma}{\lambda\rho} = \frac{0}{0*0} = Undefined$

Absolute Capillary Wave Speed in Void Bubble = $\frac{\sigma}{\lambda\rho} = \frac{P.E/Surface\ Area}{\rho=0} = \infty$

Absolute Capillary Wave Speed inside Electron Stucks = $\frac{\sigma}{\lambda\rho} = \frac{Undefined}{0*0} = \infty$

Teleportation Distance of Stucks = E/σ = Undefined/Undefined = Undefined

Why E is Undefined

P.E of Electron = mc^2 = h/λ

m = 0 & λ = 0

$0 \neq \infty$

So Energy of Entanglement is Undefined

It is important to understand the difference between these three cases... and this is precisely what the research paper described.

C - Refuting Misconception of (Universe-Extra-Universe Border) Effect

1. The Void is a Non-Medium, Not a Medium

The Extra-Universal Void (EUV) is fundamentally *not* a medium, a field, or a "space" with physical properties. It is the absolute absence of spacetime, dimensions, density, and physical constraints. Therefore, in the strictest physical sense, concepts like "velocity", "friction", and "time" are inherently undefined within it—not because they are infinite, but because they simply do not exist there.

2. CEUV: A Topological "Hole" in the Universe

When a stellar object collapses into a black hole, it does not create a massive point of singular density. Instead, it generates a Confined Extra-Universal Void (CEUV)—essentially a "bubble" of this absolute nothingness, bounded and contained by the Universe's dark matter surface tension. Think of it as tearing a "hole" in the fabric of the Universe, leading to an external, non-spatial realm.

3. Traversal is Instantaneous Topological Rewiring, Not Motion

Crucially, an object traversing a CEUV does not "travel" or "move" through it. Because there is no medium, the object experiences no drag, no Spaghettification, and no local interaction of any kind. The object does not pass through distance over time. Instead, it is **instantaneously mapped** from one region of spacetime (the black hole acting as the "gate" or inlet) directly to another region of spacetime (the "outlet" or exit point).

This is a topological rewiring of the Universe's geometry—a "shortcut" that bypasses the spatial and temporal constraints of our local reality entirely.

4. Resolving the Paradoxes of Speed and Causality

This interpretation elegantly resolves previous paradoxes without violating General Relativity or Quantum Mechanics.

- **Speed of Light:** There is no violation of the light speed limit because there is no "speed" involved. The object is not exceeding the speed of light; it is entirely outside the domain where speed is a measurable quantity.

- **Causality:** Causality is defined *within* spacetime. Since the CEUV exists *outside* our spacetime, the violation of causality is impossible. There is no "before" or "after" in the void—only an instantaneous reconnection of two separate spacetime points.

Conclusion

The framework does not rely on an "ether" or any mystical medium. It relies on a precise, geometric understanding of topological relinking. The object disappears from one point in the Universe and appears at another instantaneously, because it passes through a region where time and space are nonexistent—a pure, unconstrained void. This provides a seamless, self-consistent mechanism for black holes, entanglement, and non-local phenomena.

D - Definition of Mass/Energy/Momentum & correlation with local system entropy limits :

Mass , Energy & Momentum correlated by the formula

$(E^2) = (pc)^2 + (mc^2)^2$

As momentum is (p)

As mass is (m)

(pc) is kinetic Energy of the System studied

(mc^2) is potential energy

As Mass (M) is a form of Energy according to Special Relativity

Determining Local Entropy for Fermions , Baryons & Photons

1. Fermi Gas (μ = 0)

The local entropy for a Fermi gas with a vanishing chemical potential (μ = 0):

$$S(fermions)_{Local} = -K_B * V * [g/(2 * \pi^2 * \hbar^3)] * \int [from\ 0\ to\ \infty]p^2 * [f(p) * \ln(f(p)) + (1 - f(p)) * \ln(1 - f(p))]dp$$

$Fermi - Dirac\ Distribution\ at\ Zero\ Chemical\ Potential\ (f_0(p))$: $f(0(p)) = 1 / [e\text{^}(E_p / (K_B * T)) + 1]$ ----------(27)

2. Photon Gas

The entropy for a photon gas (bosons obeying Bose-Einstein statistics):

$$S(photon)_{Local} = -K_B * V * [g/(2 * \pi^2 * \hbar^3 * c^3)] * \int [from\ 0\ to\ \infty]E^2 * [(n(E) + 1) * \ln(n(E) + 1) - n(E) * \ln(n(E))]dE$$

$Bose - Einstein\ Distribution\ (n(E))$: $n(E) = 1 / [e\text{^}(E / (k_B * T)) - 1]$

----------(28)

3. Baryonic Gas (μ ≠ 0)

The entropy for a baryonic gas, accounting for a non-zero baryon chemical potential (μ ≠ 0):

$$S(baryon)_{Local} = -K_B * V * [\, g/(2 * \pi^2 * \hbar^3)] * \int [from\ 0\ to\ \infty] p^2 * \left[f(p) * \ln\big(f(p)\big) + \big(1 - f(p)\big) * \ln\big(1 - f(p)\big) \right] dp$$

$$Fermi - Dirac\ Distribution\ with\ Chemical\ Potential\ (f(p)):\ f(p) = 1\ /\ [\, e\text{^}(\ (E_p - \mu)\ /\ (K_B * T)\ )\ +\ 1\ ]$$

According to Proposed Theory

Theoretical Reverse Transformation from (Energy) to (Massive Particle (of Potential & Kinetic Energy) depends on Equilibrium of Local Entropy between Photons & Baryons (or Fermions)

Where

$$S(photon)_{Local} = S(baryon)_{Local}$$

& That reflects also to Leptons

Hence The Proposed Model defines quarks & leptons massive nature as intrinsic masses of VBHs

& Hadrons (including Baryons & Mesons) masses as energy of entanglement between quarks

& This perception requires & urges that retro transformation from energy to massive particle occur due to Universal inflation (global inflation) & local contraction , since at local contraction Entropy decreased so massive matter being formed , then entropy increased due to local expansion .

& This perception of local contraction & then local expansion interprets the reason of Universe isotropic characteristics

E - Universal Absolute Capillary Wave (UACW) - Light Correlation

Universal Absolute Capillary Wave (UACW) is the wave of The Universe movement itself

This wave carries (energy) by its constant speed (299,792,458 m/sec)

& Light is Energy , Energy spreads by the Speed of UACW

& that explains the constancy of speed of EM

(c) is the Universal Absolute Capillary Wave (UACW) speed that carry energy (photons)

& that doesn't mean that Photons acquire its frequency & wavelength from UACW

So , Electromagnetic wave (Energy) speed is an intrinsic nature depends on UACW according to Proposed Model

& According to formula of Universal Absolute Capillary Wave Speed which is mathematically defined as following :

$$c = \sqrt{\frac{2\pi\sigma_{UM}}{\lambda\rho}} \qquad \boldsymbol{Where\ \sigma_{UM} = \frac{E_{UM}}{A}}$$

Where UACW constituents is Universal matter (Baryons + Darkmatter) which contributes in UACW generation & defines its speed

General Relativity states that electromagnetic wave speed is constant related to all reference frames , since it has no preferred reference frame

While General Relativity fails to interpret the reason of constancy & ultimate of electromagnetic wave speed , & the reason that this concept doesn't apply in universal expansion rate , since the proposed model indicates in Beyond Universe subsection that expansion rate of the boundary of observable universe related to earths reference frame exceeds 3 times the speed of light

Well , The proposed model redefines UACW as following

UACW is Universal Absolute Capillary wave generated out of baryons movements , birth & annihilations of Blackholes as Main contributors

UACW carries all forms of energies at its speed determined by Absolute Capillary wave speed formula displayed above

Hence electromagnetic wave is - in fact - a form of energy & it has not generated mass , so it spreads at the speed of UACW which equaled (c) = 299,792,458 m/sec

Unlike General Relativity , Proposed model constitute that Absolute Capillary wave has a preferred reference frame , that frame is constituents of Universal Surface tension , which is Baryons & Darkmatter (which is defined by the proposed model as Quantum fluctuations , or as Standard Model states (Vacuum Energy))

But Dark energy (which is defined by proposed model as attraction between matter/antimatter across universe) is not within preferred reference frame of UACW , that explains UACW speed ultimate doesn't apply on Universe Expansion

The difference of matter relative movements cannot be separated from Universe Absolute Capillary wave hence every matter contribute in its generation , that explains Michelson & Morley findings , & General Relativity consideration of light speed ultimate & constancy related to all reference frames

F – Field Nature

So … According to Proposed Model as clarified earlier that

1 - All Elementary particles were point of zero mass – zero dimensions .

2 - Electrical Charges are Vector Geometry between particles entanglement & particles spin that is responsible of increasing/decreasing wavelengths .

3 - Particles masses are intrinsic potential energy of accompanied VBHs .

4 - Entanglement Energy + quarks intrinsic masses of accompanied VBH is responsible of Hadrons mass .

5 - Massive Particles is Energies that its local Entropy decreases to the amount of local entropy responsible for massive particles formation (Equilibrium Between Bose-Einstein Local Entropy Distribution & Fermi-Dirac Local Entropy Distribution) .

6 - Electromagnetic (Energy) Wave Speed is an intrinsic nature of the UACW .

Fields are generated out of Geometrical Distribution of Local Systems Entropy .

G - Casimir Effect - Darkmatter - Dark Energy

According to The Proposed Model

Casimir Effect

Casimir Effect is The Quantum Fluctuations effect

Darkmatter

Darkmatter : is The Energy of Interactions Between Matter & Antimatter at Universe Inflation Phase , Which make quantum fluctuations , as described in Universe Creation section

So Darkmatter is What is being described as The Energy of Quantum Fluctuations or The Energy of Vacuum as Standard Model call , but Unlike The Standard Model , Proposed Model represents Energy of Vacuum as Darkmatter instead of Dark Energy as Standard model suggests lately .

Dark Energy

Dark Energy : is The (Attraction) between Matter & Antimatter as it is described in Proposed Model Universe Creation Section .

As Proposed Model describe Observable Universe as local universal point in the surface area of Unlimited or Undetermined Global 4D Universal Sphere , Where matter & Antimatter survived Quantum Fluctuations annihilation being distributed across Global Universe

Matter & Antimatter attract each other through opposite entanglement &spin vectors that lead to increase in motion frequency & decrease of wavelength in between due to opposite vector geometry

Because Observable Universe is a point in Global Universe Surface , it appears locally as flat , not curved spatially .

& Because Global Universe contracting due to matter/antimatter attraction , Global Universe curvature increases , as curvature increase , **Observable Universe point in the surface appear to be expanded locally .**

This Phenomenon explain why Universal Expansion rate accelerating , while Observable universe appear flat spatially (locally) .

H - Stars Formation Stages

As Discussed earlier in Neutron/Hadron Star Lower Mass limit subsection heading

There are several lower masses depended on stellar object core dominant element (isotope)

$$M_{\text{Ch General}} = \frac{\omega_3^0\sqrt{3\pi}}{2}(\tau^2)^{\frac{3}{2}}\frac{1}{(\mu_e m_H)^2} / \frac{\text{mass number of sigma mesons \& } neutron\ mixture}{\text{mass number of dominant core isotope}}$$

$$M_{\text{Ch General}} = \frac{\omega_3^0\sqrt{3\pi}}{2}(\tau^2)^{3/2}\frac{1}{(\mu_e m_H)^2} * \frac{\text{mass number of dominant core isotope}}{\text{mass number of sigma mesons \& } neutron\ mixture}$$

$$M_{\text{Ch General}} = \frac{\omega_3^0\sqrt{3\pi}}{2}(\tau^2)^{3/2}\frac{1}{(\mu_e m_H)^2} * \frac{(1.67\ \text{Mass number})}{(0.867\ \text{Atomic Number}) + (1.67\ \text{Neutron Number})}$$

$$M_{\text{Ch General}} = \frac{\omega_3^0\sqrt{3\pi}}{2}(\tau^2)^{\frac{3}{2}}\frac{1}{(\mu_e m_H)^2} * \exists$$

Where

$\tau = 0.5\sqrt{\frac{\hbar c}{G}}$ = Corrected Planck Mass = 1.088217*10^-8 kg

ω_3^0 is Lane Emden Polytrope Constant $= 2.018236$

m_H is Mass of Hydrogen Atom (Mass of Nucleon) $= 1.67 * 10^{-27}$ kg

$$\mu_e_{Iron} = \frac{\left(\frac{0.867}{1.67} * 26\right) + 30}{26 * 1.67/0.867} = 0.86856155032$$

$$\exists_{\text{Iron}} = 1/\frac{(0.867*26) + (1.67*30)}{55.845*1.67} = 1/\ 0.77890954593 = 1.28384612209$$

$$M_{\text{Ch Iron}} = 2.43614349808 * 10^{30} = 1.2251346503\ Solar\ Mass$$

$$\mu_e_{Silicon} = \frac{\left(\frac{0.867}{1.67} * 14\right) + 14}{14 * 1.67/0.867} = 0.78869052315$$

$$\exists_{\text{Silicon}} = 1/\frac{(0.867 * 14) + (1.67 * 14)}{28.085 * 1.67} = 1.32051213469$$

$$M_{\text{Ch Silicon}} = 3.0389268362 * 10^{30} = 1.52827391723\ \text{Solar Mass}$$

$$\mu_e_{Oxygen} = \frac{\left(\frac{0.867}{1.67} * 8\right) + 8}{8 * 1.67/0.867} = 0.78869052315$$

$$\exists_{\text{Oxygen}} = 1/\frac{(0.867 * 8) + (1.67 * 8)}{15.9994 * 1.67} = 1.31646620024$$

$$M_{\text{Ch}_{\text{Oxygen}}} = 3.02961582841 * 10^{30} = 1.52359141873 \text{ Solar Mass}$$

These are Ultimate masses of Stellar Objects Of different dominant element core begins with Oxygen till Iron , where PEP for its electrons are conserved before being violated due to excessive mass/gravity gain

Star formed out of Hydrogen Dust gathered through gravity … gravity on star generated Hydrogen fusion into helium , after that helium into carbon , carbon into oxygen , oxygen into silicon , silicon into iron (dead end)

Forming element is dependent on the star initial mass
While formation of White dwarfs , Neutron/Hadron Star & Blackholes is independent from that stage , these 3 scenarios dependent on the mass of star remnant after its initial mass is not enough for element on its core to fuse

Neutron/Hadron Star can be formed of iron if the remnant of the initial star exceeds 1.22:1.23 solar mass , as its Initial Star was massive enough that its core elements produce iron out of nuclear fusion

Neutron/Hadron Star cannot be formed of Oxygen or Silicon if the remnant of the initial star fall short of 1.52 solar mass , as its initial star wasn't massive enough to fuse Oxygen & Silicon into Iron

There will be argument that there were Observation of HESS J1731–347 neutron star of mass 0.77 solar , which violates the creation of sigma mesons phenomena of Hadron stars

The 1st Probability …. this observation is suffers huge uncertainty , that is confidence level cannot surpass 68.3% (1 sigma) , so , the observation is not reliable

The 2nd Probability …. The Neutron/Hadron Observed is in its mid or near end of its decay process … so its mass decrease over time … & this means its an old formed Hadron/Neutron star

I - Blackhole Formation

After clarifying Proposed model description of various scale physical occurrences , Proposed model deduce Blackhole formation as follows :

Blackhole forms out of Massive star , that its mass magnitude allow the star to fuse Hydrogen , to helium , to carbon to oxygen , to silicon all the way down to iron

Iron core is able to resist rest mass of a star due to its atomic & nucleus stability , Pressure rebound occurs

Star explodes in a Supernova

1.85 Solar mass is enough to overcome strong interaction , which is triple entanglement of The stellar Object remnant

But that magnitude mass limit increases to 2.37 Solar mass , in order to overcome specific weak interaction , that of interaction of overcoming PEP for electrons to fuse with protons creating sigma mesons & stabilizing sigma mesons continuous decay by continuous formation

If The star exceeds 2.37 solar mass , it will contract all neutrons as their wavelengths decreases & frequency increases until their wave length reaches Planck length of 1.616*10^-35 m & velocity of 0.999999 c & their mass (strong interaction) reaches 1.09*10^-8 kg , then surpassing that limit (reaching speed of light) will do two things

In microscopic scale , it annihilate strong interaction of 3-way entanglement for every neutron (& equivalent 4-way entanglement for sigma mesons) , freeing all point charges quarks that disruption energy rebound of entanglement annihilation destroys darkmatter region until its surface tension stabilize that rebound , forming a CEUV , that works as a topological shortcut between coordinates of Schwarzschild sphere & CEUV sphere ,where blackhole is its inlet & CEUV its outlet

In macroscopic scale , escape velocity of the stellar object remnant after contraction reaches speed of light , meaning light cannot escape gravity of the blackhole

Past Time Dragging (PTD) Phenomenon - Quantum Information Conservation

Different magnitude of expansion rate affect Universal Age - Baryons/Darkmatter Ratio

For BBN model

Assume Hubble Expansion rate to increase to be of magnitude of 73km/sec/megaparsec

Megaparsec = 3.085677922066030760637920417 89 * 10^22 m

Hubble expansion rate (H) = 73000/(3.085677922066030760637920417 89 * 10^22)

= 2.36576862018*10^(-18) Hz

Universal Critical Density (Cr) = $\frac{3H^2}{8\pi G}$

Where H = 2.36576862018*10^(-18) Hz

G = 6.6743 * 10^-11 m^3 kg^{-1} sec^{-2}

Cr = 1.00096802671*10^(-26) kg/m3

Where (h) = 0.73

h2 = 0.5329

$\rho_{critical}$ = Cr/h2 (gm/cm3) = 1.87834120231*10^(-29) gm/cm3

(& That magnitude would answer questioning validity of the mechanism of Universal global contraction & local expansion , hence according to Friedmann formula , density directly proportional to Hubble expansion rate squared , if accelerating expansion is being fostered , this occurrence cannot take place unless Universe is contracting globally , not expanding)

From BBN model

Baryon Mass Density = $\rho_{BM} = 4.1873187451 * 10^{-31}\ gm/cm^3$

$\Omega_{BM} = \frac{\rho_{BM}}{\rho_{critical}}$ $=(4.1873187451 * 10^{-31})/(1.87834120231 * 10^{-29}) = 0.02229264172/h2$

Baryon matter ratio = 0.02229264172/0.5329 = 0.041832692287483580409082379 43329

= 4.183269228748358040908237943329 %

Darkmatter ratio = 0.31830988618 - 0.041832692287483580409082379433 29 = 0.27647719389 = 27.647719389 %

Universal Age = t© =

$$\frac{1}{2.36576862018 * 10^{-18}} * \int_0^1 \frac{dz}{z\sqrt{(0.00009)(z)^{-4} + \left(\frac{1}{\pi}\right)(z)^{-3} + (0)(z)^{-2} + (0.68160011382)}}$$

= 4.00638393498652107901508398 81*10^17 sec = 12.704160118551880641 22 billion years

Proposed Model Vision of Hubble expansion rate - Past Time Dragging (PTD)

Proposed Model fostering CMB method as the current magnitude it has been mentioned before in framework layout section , but views Hubble expansion rate different than standard model

According to Proposed Model & to General Relativity

Velocity contribute to Space & time dimensions (Proposed model focuses on time dimension)

at (Speed of Light) according to GR (at Universal Absolute Capillary Wave (UACW) speed according to Proposed Model) Spatial dimensions contract to zero & time dimension extends to infinity

While according to GR assumes Light speed as axiom

Proposed Model derivate light speed as energy transmitted by UACW speed by magnitude of (c) , which Baryon & Darkmatter contribute to UACW , so UACWS (Light speed) is constant to all matter & darkmatter references , but Dark Energy doesn't abide , which is explained earlier

While Dark Energy doesn't abide by UACW , it does not abide by time too

Actually , Dark energy violate speed limit Space & time
According to Proposed Model , that Dark Energy is the attraction between matter & antimatter , displayed by Hubble expansion rate

According to Proposed model

As Hubble expansion rate accelerates , it means that The Universe get (Younger) as it is introduced mathematically before

Meaning Past is being (pulled) towards present

So when Hubble expansion rate is observed 67.4 km/sec/mpc & universe age as 13.76 Bil years is correct

& when Hubble expansion rate will accelerate to 73 km/sec/mpc , so Universe (will be aged) at 12.7 billion years

(this Hubble expansion rate magnitude used in example doesn't relate to Standard Candle model , as Standard Candle observes (Evolution) of expansion rate , not the current expansion rate .

That means at the end of universe , where matter & antimatter fused , The Past of The Universe , & Present became compatible .

Proposed model extends time manipulation of General Relativity to the extremes

That occurrence that Proposed model explained , forms the extreme application of conserving Quantum information

As Past being fused to present

No information can be hidden nor ignored , nor neglected , nor denied , nor forgotten

Because it is being applied & displayed in present

Past become Present

That what (Past Time Dragging) - PTD - means .

IX) Problems of Hawking Radiation & Solutions :

Primary Version

1) Hawking radiation interpretation demonstrate gradual reduction of the mass and rotational energy of black holes and consequently cause black hole evaporation. Because of this, black holes that do not gain mass through other means are expected to shrink and ultimately vanish. For all stellar blackholes except the smallest ones, this happens extremely slowly. The radiation temperature, called Hawking temperature, is inversely proportional to the black hole's mass, so micro black holes are predicted to be larger emitters of radiation than larger black holes and should dissipate faster per their mass. Consequently, if small black holes exist, as permitted by the hypothesis of primordial black holes, they will lose mass more rapidly as they shrink, leading to a final cataclysm of high energy radiation alone. Such radiation bursts have not yet been detected. (29)

this radiation is black-body radiation released outside a black hole's event horizon due to quantum effects according to a model developed by Stephen Hawking in 1974.[1] The radiation was not predicted by previous models which assumed that once electromagnetic radiation is inside the event horizon, it cannot escape. Hawking radiation is predicted to be extremely faint and is many orders of magnitude below the current best telescopes' detecting ability.

Hence Hawking temperature equation

TH = hbar.(C^3)/(8pi.G.M.K)

Where K is Boltzmann Constant

But the main concern & struggle with Hawking radiation & his assumption of Blackhole inevitable evaporation model . is the nature of blackhole matter that assumed to be dissipate under heating radiation

According to relativistic common assumption of blackhole , that its matter is concentrated in a singularity , if we consider hawking radiation , that radiation will not affect blackhole event horizon until the total annihilation of blackhole core matter , event horizon will suddenly vanish

Hawking radiation , cannot explain physically how could a blackhole would evaporate , Hawking theory did not consider that radiation would be attracted to blackhole core & fused

with its matter again due to enormous amount of gravity applied , so the evaporation will never occur .

To demonstrate this problem clearly

The proposed model does not reject the existence of Hawking radiation in principle; the difference lies only in its characterization. The radiation is produced during the formation of the black hole (as a result of heat exchange between compressed neutrons and outer space), and not as a byproduct of the black hole itself. Furthermore, it has no effect on the fate of the black hole or its evaporation, because the black hole no longer has a physical body, but rather is the gateway to a bubble of Extra-Universal-Void in three-dimensional space.

Additionally, Hawking radiation is a form of energy. If this energy is at the event horizon, as in the prevailing relativity model of black holes as point particles, then Hawking radiation has no effect on the evaporation of the black hole. This is because the generally accepted point particles are unaffected by the radiation. Even if a point particle were attracted to the black hole, the black hole's gravity would quantize and deflect the radiation's energy until it becomes another point particle within the black hole's matter. In other words, the black hole's mass would increase (it would not evaporate).

Therefore, the prevailing point-particle model of the black hole, as relativity assumes, violates the principle of Thermodynamics, and also Hawking's interpretation if adopted, violates relativity in order to conform to thermodynamics; therefore, Hawking's interpretation does not solve the problem, but rather jumps over it.

So , The Only possible explanatory interpretation of Hawking Radiation , is that star surface neutron as its wavelength reduced due to increasing Kinetic Energy due to Gravitational Crush

Its Temperature increases according to Wien’s & Planck’s Law , Surface Heat Transfer to surrounding area which turned into border of Event Horizon that the radiation cannot escape

This occurrence - as interpreted - won’t cause Blackhole Evaporation
(which became not a matter anymore) , it doesn’t affect it

Advanced Version of Hawking Radiation

2) Even The Advanced version of Hawking Radiation to compensate for Original model fallacy that assumes that any particles that Blackhole accrete generating two entangled particles , one of them is accreted to Blackhole core , the other radiates at event horizon , displaying the accreted partner of having negative energy that will reduces Blackhole mass

Even this idea has much more fallacies than original one

This new version contradicts standard model it defends , because according to standard model , potential energies (such as that of Blackhole itself) considered a negative energy , so the accreted particle increase Blackhole mass (according to Standard model) , not reducing it

& virtual particles of negative mass is an unproven phenomenon , hence matter & antimatter themselves both are affected by their masses & gravitational field the same way

& a Hypothetical such virtual particles of negative mass , it will interact gravitationally inversely , so it will never be accreted by Blackhole

Casimir effect is considered negative energy because it is (shrinking effect)

& even if Hawking radiation being approved , the information of the radiation will be wasted

& that contradicts with Quantum information conservativity principle

Which will lead to violation of Energy conservation principle

So displaying a negative energy in nature (even to virtual particle) is an ad-hoc mathematical fallacy) , & it doesn't solve any problem .

SO : in conclusion , Thermodynamic equilibrium is set , Hawking Radiation has no relation to Blackhole falsely claimed evaporation , Blackhole is just gate of Universal Topological Shortcut

Hawking Radiation cannot be detected neither outside of event horizon

& Experimental Acoustic Blackhole is just hypothetical analogy of cosmic Blackhole , which won't prove anything due to different conditions , & even in these circumstance , Hawking analogous radiation does not affect the analogous acoustic blackhole .

Conclusion

3) So , Considering Hawking Radiation as Energy Emission of Blackhole that will lead to The inevitable Evaporation of Blackhole is Just an Ad-Hoc due to following reason :

1 - to compensate for Thermodynamics , It violates General Relativity

2 - to evade violation of GR , it violates causality , where Hawking radiation assume that the radiation evaporate blackhole from event horizon outside , while Event Horizon is Just a space where remnant of collapsed star were , before collapsing into singularity inside the blackhole (according to Standard Model) - So Event Horizon is Just a Space for Gravitational Effective field of Center of Mass (Actually the total mass) in the singularity inside blackhole (Again According to SM) & Hawking Radiation won't affect Blackhole by any means … its effect on SM Blackhole is the same effect as killing some enemy by stabbing his (shade) of his body on the floor .

3 - Advanced Versions of Hawking Radiation are not better solution either to the original form , but may be much worse versions , manipulating math & ignoring physics of it & violate QM first principles , & it doesn't solve any problems .

4 - & no need to inform That SM Blackhole itself violates Entropy entirely

5 - This critic does not criticize Hawking Radiation as an occurrence , but discuss its cause & questions its reasons & physics behind it & question its fate .

X) Predictions :

Cosmic

1) Teleportation Distance Predictions

Blackhole Stellar Object	Mass (Solar mass)	Schwarzschild Radius (SI)	CEUV Radius (LY)
Minimal theoretical Blackhole (PFRGSQD)	2.372676974	7002.805797	0.112
GW190814's Secondary Object	2.60	7673.735308	0.118
The Unicorn (V723 Mon)	3.00	8905.651049	0.127
XTE J1650-500	3.80	11280.49133	0.143
GRO J0422+32	4.20	12467.91147	0.150
NGC 3201-1	4.36	12942.87952	0.153
GAIA BH01	9.62	28557.45436	0.227
Sagittarius A	4,200,000	12467911468	149.859
Messier M87	6,000,000,000	1.78113E+13	5,664.144
Ton 618	66,000,000,000	1.95924E+14	18,785.840
Phoenix A	100,000,000,000	2.96855E+14	23,123.770

2) Blackhole Formation Limit

As it was previously concluded in Hadron Star subsection

The Maximum Mass Hadron Star can be formed = Minimum Mass a Blackhole can be formed

= 1.848100744828204785123057044785 9 /0.77890954593

= 2.37267697447 Solar Mass

3) Formation of Blackholes (The Birth of Blackholes)

Proposed model predict that Particles accreted are being teleported in all directions surrounding the stellar core at all distances till the surface particles being teleported at CEUV distance of the stellar object formed into a blackhole

Byproduct particles , which were quarks of different charges due to being previously formed out of neutrons

Muon & Anti Muon with corresponding Muon Neutrino/Anti Muon Neutrino Counterparts 99.8% occurrence

electron/positron with corresponding electron neutrino/anti neutrino counterparts 0.2% occurrence

Gamma rays 98.8% occurrence

Or in electron + positron + gamma rays 1.2% occurrence

This constitutes in the end of the final constituents

Protons + electrons + Gamma rays

But all of teleported particles scattered at CEUV circumference .

4) Neutron/Hadron Star

Due to Sigma Meson mass range from 0.713*10^-27 kg to 0.98*10^-27 kg (30)
Chandrasekhar Limit of Neutron Star will range from

0.68942125753 solar mass @0.980*10^-27 kg (550 MeV meson)

1.57921876676 solar mass @0.713*10^-27 kg (400 MeV meson)

While The Framework locked the magnitude of sigma meson @0.867*10^-27 (486.35 MeV) ------- (25)

Making Chandrasekhar mass limit yields from 1.22 to 1.53 solar mass … according the star core element at its final stage

Energy Magnitude from Neutron Star formation due to Protons/Mesons Mass loss

= 4.9015382 * 10^46 joules

Microscopic

Entanglement Efficiency

derived from Wien formula (20)

T*$\lambda = 0.0029$

As

T=0

$$\lambda = \frac{0.0029}{0} = \infty$$

As system energy = h/ λ = h/∞ = zero

Atoms Quantum States emit energy (As of Hydrogen for example)

Quantum State Energy = 10.2eV = 1.63422*10^-18 joules

Its wavelength $\lambda = \frac{hc}{E} = 6.62607015 \times 10^{-34} * 299792458/1.63422 * 10\text{^} - 18$

= 1.2155 * 10^-7 m

That is the amount of energy that Quantum State emit in a system (Theoretically) reach 0 Kelvin to balance the surrounding of reaching (Zero Kelvin) , Which is equivalent to the amount of energy that the electron is stabilized in its quantum state , where occurrences such as (Superfluidity , Superconductivity , Entanglement) reach the efficiency of 100% , so emitted energy creating extra (VBH) Equivalent to stable quantum state of Hydrogen Atom without excitation

As Entanglement Control Parameter Efficiency (η) = 1

Computing the temperature where entanglement has (Zero Efficiency – Uncontrollable entanglement)

Where T*$\lambda = 0.0029$

T = 0.0029/ λ

= 0.0029/(1.2155 * 10^-7) = 23,858 Kelvin

(η) = 0

But the Efficiency Curve isn't linear

According to Friedmann Equation , Radiation ratio is (0.00009) & parameter of ((a)^4)

(η) = 0 T = 23,858 Radiation ratio = 0.00009 (Rad parameter) = ((a)^4)

(η) = 1 T = 0 Radiation ratio = 0.00009 (Rad parameter) = ((a)^4)

Where X = (η) , U (is Constant)

$$T(\eta) = \frac{U}{(0.00009a^4)^n}\left[\frac{e^{-\frac{\eta}{(0.00009a^4)^n}} - e^{-\frac{1}{(0.00009a^4)^n}}}{1 - e^{-\frac{1}{(0.00009a^4)^n}}}\right]$$

As (n) is element electron quantum states occupied

Forward Prediction Applied

3) Observational comparisons presented here are forward predictions derived from independently fixed parameters. No free parameter fitting or post-selection has been employed."

XI) Suggested Experimental verification of the proposed theory

Verifying that model of CEUV principle at cosmic level is a must , in order to verify the interpretation at microscopic scale

Natural

a) Observe a small blackhole near earth (within our detectors reach , where suggested distance of its Void teleportation distance is minimal to be able to observe any occurrences happen in such narrow range spectrum , observe any objects appear , or any blue-shift spike appear in its proximity (On determined surface area of the predicted Void bubble) , providing ability to narrow the search by predicting endpoint of Void bubble radius coords hence Outlet vector is an extension of inlet vector

Artificial

b) Observing Controllable Entanglement in Quantum Computing application in environment near zero Kelvin , where entanglement efficiency grows as it we reach closer to Zero Kelvin .

c) Feasibility to make Artificial Blackhole is not impossible .

Blackhole won't be made by reaching frequency equivalent to overcome Dark-matter Surface tension , hence it will need frequency exceeds 10 million times than Planck Frequency .
the other method to make an artificial blackhole is reaching the (Universal Resonance) to form a desired blackhole with enough amplitude to compensate for the desired object to be teleported .

The only way to reach the resonance frequency , is to determine the Percussion Wavelength Of the universe as mentioned earlier in the proposed frame work which is equivalent to 2/3 Comoving Distance of the Universe .

& This doesn't need to reach such a very low frequency which will force the performer to wait billions of years waiting for such a frequency .
Reaching Resonance frequency only require a Harmonic Wave which its node compatible to the Universal Resonance Frequency .

The Only Technical Obstacle is to determine the exact Radiation Energy Ratio to determine Dark Energy , in order to calculate current Comoving Distance needed to know to deduce Center of Percussion to determine Resonance frequency .

But this Technical Problem solved , by applying mechanism of using Harmonic Wave Frequency Which is known to be larger than Universal Resonance Frequency , then applying Lower Frequent Harmonic Wave , then applying trial & error mechanism to narrow the proximity of The Resonance Frequency , Until Resonance Frequency is marked , amplified enough for the desired object to be teleported .

C) Conclusion :
Blackholes Cannot be formed naturally unless it's being Virtual Blackholes (Beyond Planck Limits) , Or Cosmic Blackholes which is minimum mass equals 2.37267697447 Solar Mass

The Rest only could be made artificially

XII) Promising Observations Validating Theory

1) Scattering Experimental measuring Proton Charging Radius is 0.8409 fm (While Proposed Framework Calculation is 0.8412 fm (measuring Octants) - 99.9% Precision - Solving Proton Charging Radius Paradox . (22)

2) Blueshift Spikes are being Observed at sub 6000 light years away from M87 Messier Blackhole , recent analysis conclude that spikes occurs at range of 5500-5700 Light years , that matches Proposed Framework predictions of Void Bubble Radius of M87 at 5664 Light years - 99.7 (+ or - 1%) Precision . (31)

3) Strange Non-thermal Radio Filaments are being Observed in Sub 150 Light years away from Sagittarius A Blackhole , which Framework Void Bubble Radius Prediction Locked in 149.859 Light year radius (+95% precision) . (32)

4) Strange Cooling Filaments are being Observed at range of 20,000 - 30,000 Light years away from Phoenix A Beast Blackhole , (Proposed Model prediction Locked in 23,123 light years Bubble Radius . (33)

5) Most massive White Dwarf detected & Observed is (ZTF J1901+1458 with announced mass of 1.346 ± 0.019 solar mass ($\pm$1.4 % margin of error) (34)

with combining gravitational redshift (how much light is stretched by the star's immense gravity) with modern atmospheric modeling yields average mass precision errors within 6% **,** adding up margin of error up to 7.4% (35)

Even with perfect data, theoretical models introduce a base layer of systematic error:

Subtle variations in the expected internal profile alter how dense the star is, which adds an intrinsic **1% to 3% systematic margin of error** to mass calculations derived from radius measurements (36)

XIII) Hints

1) proposed model doesn't resolve the physical refusing paradox of (Fine Tuning) ... as discussed model doesn't consider Fine Tuning as a physical problem , in fact , the model consider Fine Tuning is essential & unshakable proof , that is not related to any metaphysical principles ... but in rigorous mathematical & physical principles & terms

Fine tuning is Essential in material formation in standard model as

solving matter-antimatter paradox as the inevitable formation relying into scenarios

Scenario Number 1 (proposed framework recommendation) :

Big Bang explosion is so powerful , that dark energy splits matter & antimatter to the extreme far sides of universe , that quantum fluctuations if happened in one side to make matter-antimatter instability to occur , it should be occur in all sides symmetrically or asymmetrically , while power of explosion & range of distributions of matter & antimatter prevents complete annihilation of forming matter & antimatter

but explosion at that power magnitude & that proximity , isn't essential to occur , This meant That magnitude & proximity should selection of Fine tuning to be occurred in first place .

it displays selection of one magnitude probability out of (infinite) magnitudes , & that cannot be interpreted other than inevitable fine tuning & controlled selection , & cannot be a coincidence mathematically , hence $1/\infty$ = zero

Scenario Number 2 (Standard Model) :

magnitude of Big Bang is only enough to occur. but not enough to overcome matter-antimatter total annihilation through stability , that model needs quantum fluctuations to preserve the forming of the matter safely as we see , quantum fluctuations & these fluctuations magnitude should also be as selection of Fine tuning to occur .

Baryonic matter ratio of 4.90% , cannot be applied without fine tuning

also in Quantum level , also fine tuning is essential in Proposed Model .

- Quantum levels & VBHs existence in it which its accumulative mass measures for handling Particles precise intrinsic mass & Entanglement precise generated energy

that would not happen without fine tuning

so , Fine tuning in both scenario is inescapable fact , without any metaphysical term applied

& this model , or concept if it is tuned & verified , doesn't need multiverses that formed infinitely in order to unify Relativity with Quantum Mechanics

Why This Framework claimed to be Complete by our Current Standards

This framework is claimed to be complete **by necessity of its construction**, not by accumulation of auxiliary formalisms or extensions. Its completeness does not rely on mathematical fashion, institutional standards, or external closure principles, but on **mechanical and ontological closure**.

---

1. Completeness by Single Ontology

The framework claimed that is founded on **one physical substrate** with well-defined mechanical properties (density, surface tension, resonant length).
There is no ontological split between:

- spacetime and matter,
- quantum and classical domains,
- particles and fields,
- information and dynamics.

All physical phenomena arise from **the same medium governed by the same rules**.
A theory that does not require additional entities at different scales is complete at the ontological level.

---

2. Completeness by Mechanical Closure

The framework specifies:

- what exists (the medium),
- how it responds to energy (curvature, tension, void formation),
- how motion, limits, and stability arise,
- and how breakdowns (black holes, quantum transitions) occur.

There is **no missing mechanism** that must be supplied externally.
Every phenomenon is generated by mechanical response, not postulated behavior. This constitutes **mechanical closure**.

---

3. Completeness by Necessary Fine-Tuning

The framework does **not avoid fine-tuning**.
It demonstrates that fine-tuning is **structurally mandatory**.

The universe can only exist at precise equilibrium and resonance conditions.
Any deviation leads not to a different universe, but to **mechanical non-existence** (instability, collapse, dispersion).

Thus, all observed numerical relations are:

- not adjustable,
- not probabilistic,
- not anthropic,
- but mechanically enforced.

A theory that explains *why values must be what they are* is complete in its explanatory scope.

---

4. Completeness Across Scales Without Regime Change

The same principles apply at:

- cosmological scales (expansion, dark energy),
- astrophysical scales (black holes, accretion),
- microscopic scales (orbitals, tunneling),
- informational scales (entanglement, nonlocality).

There is **no transition point** where new laws are introduced.
Scale change does not imply theory change.

This absence of regime fragmentation is a defining mark of completeness.

---

5. Completeness by Predictive Determinacy

The framework produces **non-tunable quantitative predictions**, including:

- universal matter ratios,
- black hole mass thresholds,
- void bubble radii,
- Hadron Star minimum & maximum mass limit,
- proton charge radius.

These values follow necessarily from the internal structure of the theory.
They cannot be independently altered without breaking consistency.

A theory that yields fixed outcomes rather than adjustable fits is predictively closed.

---

6. Completeness Independent of Formalism

Lagrangian , fields, and quantization schemes are **representational tools**, not criteria of completeness.

This framework operates at a **pre-formal mechanical level**, where physical necessity precedes mathematical encoding.

A theory is complete when nothing physical is left unexplained — not when every phenomenon is written in a preferred formal language.

---

7. What Completeness Does *Not* Mean

Completeness here does **not** mean:

- institutional acceptance,
- computational convenience,
- or conformity to existing frameworks.

It means:

- no missing physics,
- no unresolved sector,
- no external explanatory dependence.
- no conflict with proven observation

Conclusion

This framework claimed to be complete because:

1. It has a single ontology.
2. It has mechanical closure.
3. It embeds fine-tuning as necessity.
4. It applies universally across scales.
5. It produces non-adjustable predictions.
6. It leaves no phenomenon unexplained.
7. It requires no external theoretical scaffolding.

Nothing needs to be added.
Nothing can be removed without collapse.

The framework is complete **because it cannot be otherwise**.

Facing Physical Judgement & Concerns

Concern (1):

The work lacks a Lagrangian or field equations and therefore cannot be considered a physical theory.

Response:

This is a **methodological bias**, not a scientific argument.

Physics is not defined by Lagrangians, but by:

- explanatory power
- falsifiable predictions

The framework provides a **pre-field physical mechanism**, explaining why field theories work as effective limits.
Both QM and GR historically preceded their final variational formulations.

Requiring Lagrangian & field equations is an **ontological error & fallacy**, & decision of not using them is not a deficiency , but it is a correct & logical decision.
it is a conceptual error of demanding field expression to pre-field framework in first place

We're talking about states that work as (field generator) as the title of this paperwork indicates, & The entire framework & proposed model apply.

The use of field formulae for both quantum theory and relativity is the real obstacle to reconciling relativity and quantum mechanics in extreme states (such as blackhole & quantum entanglement) .

The mechanisms for calculating gravitational and quantum fields, as emerging phenomena, cannot explain the natural phenomena that create relativistic and quantum phenomena's in the first place. This is the paradox that has plagued attempts to reconcile relativity and quantum mechanics for so long. Therefore, resorting to an algebraic formula to unify relativity and quantum mechanics was the optimal approach according to the proposed framework.

Concern (2):

Introducing a universal medium revives the experimentally rejected ether concept.

Response:

This objection relies on a **historical misconception**.

- Classical ether:
    - defines a preferred frame that violate Michelson & Morley findings
- The proposed model:
    - defines preferred frame that doesn't violate Michelson & Morley findings
    - generates relativity itself

So , This is not a revival of classical ether anymore , but an Interpretation of Why electromagnetic wave speed is constant in all scales

---

Concern (3):

Infinite speed inside the Void violates causality.

Response:

Causality is only defined **within the medium**.

The Void:

- has no spacetime
- no metric
- no trajectories
- no velocity

The process is not motion but a **topological shortcut**.
Violation of causality is impossible where causality is undefined.

Concern (4):

Linking atomic stability to a cosmic equilibrium spatial collapse mechanism is speculative and unjustified.

Response:

This is an **intuitive discomfort**, not a logical refutation.

In the framework:

- atoms are not closed systems
- stability is an emergent balance of substrate recovery

The same mechanism operates across scales.
Different scales do not imply different physics.

Concern (5):

A "theory of everything" is inherently unscientific.

Response:

This is a **sociological objection**, not a physical one.

Science does not reject scope.
It evaluates:

- consistency
- falsifiability

The framework satisfies both versions.

Concern (6):

The manuscript arbitrarily redefines fundamental concepts and should therefore be dismissed.

Response:

Scientific progress requires redefining concepts when existing definitions fail.
The framework explains why standard definitions emerge as effective limits.

Conceptual conservatism is not a scientific criterion.

Concern (7):

The work is philosophical rather than quantitative.

Response:

The framework contains quantitative predictions and numerical estimates.
Explanatory depth does not negate scientific validity.

Concern (8):

The Void is unobservable and therefore unscientific.

Response:

Many accepted entities in physics are indirectly inferred.
The Void is defined through its observable consequences.

Concern (9):

Cross-scale unification is unjustified.

Response:

Physics historically unifies scales; separation is artificial.

Concern (10):

Predictions are adjustable.

Response:

The framework defines a single falsification point.
Failure cannot be patched.

Plus , How Predictions could be adjustable , if it predicts occurrences by particular magnitude that are unknown to scientific community till date of publishing this framework

---

Concern (11):

The work is disconnected from current literature.

Response:

Truth is not a function of citation density.

Concern (12):

The manuscript assumes a physical "substrate," which effectively revives ether-like concepts and lacks experimental justification.

Response:

The Traditional concept of ether is not existed in this proposed model , as all matters in universe contribute in forming reference frame , as it was discussed in detail in conclusion section & also interprets dark energy violation of Light speed constancy in Special Relativity

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Concern (13):

The manuscript assumes a physical "substrate," which has constant physical characteristics such as Surface Tension & Density , doesn't that contradict with formation of intergalactic cluster HALOs , Which matters gathered at more dense Darkmatter , & if so , How's variable dense darkmatter allow for constant Speed of light .

Response:
Speed of Light Constancy Originated from the Capillary Wave (Absolute) Speed which occur at Center of Percussion at Resonance state of the physical medium , that meant variable dense region won't affect results

Resonance frequency is constant for complex Object , this concern resulted from the mix of old ether phenomenon & substrate phenomenon

Plus Darkmatter explained in proposed model as the energy generated from Matter/Antimatter interaction (Quantum Fluctuations)

Plus The Concept of variable densities of Dark matter is being refuted in this framework , as it claims variable dense as primordial state of dark matter in universe creation phase

So Concern is Answered from the Framework itself .

Finale

Finally

This is proposed framework , laid down to be revised analyzed & validated constructively totalistically to greatest attention to detail , expressed in such more rigorous mathematical form , while revising proposed model considering the current theories are rigorous & tested profoundly , but they are also failed at specific ranges , that make them like all their previous counterparts , are perfectly precisely approximation works elegantly at common range , but hurdles at the extreme conditions & cases , even this framework if it was validated & verified , it will be (eventually) replaced by more accurate descriptive theory , & vises versa

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